문제를 완결성있되 가능한 한 구체적으로 서술한다.
이 Page에선 뒤로가기, 선택 지우기 등을 금한다.

「2018. Fall.

1. Let $A \in M_{n\times n}(\mathbb{R})$ such that $A^2=0$. Prove
$$2\cdot \operatorname{rank} A \le n.$$

Proof. Given $x \in \mathbb{R}^n$, $A(Ax) = A^2 x = 0$, so $\operatorname{Im} A \subseteq \ker A$. So, $\dim \operatorname{Im} A \le \dim \ker A$.
By the dimension theorem,
$$n = \operatorname{nullity} A + \operatorname{rank} A \ge 2\operatorname{rank} A$$
$\blacksquare$

2. Determine over $\mathbb{R}$,
$$A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$$
is diagonalizable.

Proof. The characterstic polynomial of A is
$$\det\begin{pmatrix} t & 0 & 2 \\ -1 & t-2 & -1 \\ -1 & 0 & t-3 \end{pmatrix} = t(t-2)(t-3) + 2\cdot(t-2) = (t-2)(t^2-3t+2) = (t-1)(t-2)(t-2)$$

By the Cayley-Hamilton theorem, the minimal polynomial has same roots to charaterstic polynomial, so we need to check that $(A-I)(A-2I)=0$,
$$\begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}\begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0$$

So, the minimal polynomial of A is $(t-1)(t-2)$, it implies A is diagonalizable, because the minimal polynomial splits over $\mathbb{R}$ and separable. Explicity,
$$\begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \iff \begin{cases} x = -2z \\ x+y+z=0 \end{cases} \Rightarrow \ker(A-I) = \operatorname{Span}\left\{\begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}\right\}$$
$$\begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \iff \{ x+z=0 \Rightarrow \ker(A-2I) = \operatorname{Span}\left\{\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}\right\},$$

Therefore, $A = UDU^{-1}$ where $U = \begin{pmatrix} -2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$ and $D = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$.

[2018. Fall,

3. Let R be a commutative Ring and let $x \in R$ such that $x^{2018} = 0$. prove that $1+x$ is unit.

Proof. observe that

$$(1+x)\cdot(-x^{2017}+x^{2016}-x^{2015}+\cdots-x+1)$$

$$= 1-x+x^2+\cdots+x^{2016}-x^{2017}$$
$$+x-x^2+\cdots-x^{2016}+x^{2017}-x^{2018}$$

$= 1$, since $x^{2018}=0$. So, $1-x+\cdots+x^{2016}-x^{2017}$ is inverse of $1+x$.

4. Prove that A_4 does not have a subgroup of order 6.

Proof. Since $|S_4 : A_4| = 2$, $|A_4| = 4\cdot 3 = 12$.
Suppose that A_4 has a subgroup H of order 6.
Then, $|A_4 : H| = 12/6 = 2$, So $gH = Hg$ for any $g \in A_4$, it means $H \trianglelefteq A_4$ is normal.
Consider the afforded Homomorphism of A_4 acts A_4/H as left multiplication:

$$\varphi : A_4 \to S_{A_4/H} ; g \mapsto (xH \mapsto gxH)$$

And, Since every 3-cycle $\sigma \in A_4$ satisfies

$$(\varphi(\sigma))^3 = \varphi(\sigma^3) = \varphi(id) = id$$

and $S_{A_4/H} \cong \mathbb{Z}/2\mathbb{Z}$ So $\varphi(\sigma) = id$. It implies all 3-cycles are contained in $\ker\varphi$. Meanwhile, observe that $(1\,2\,3), (1\,2\,4) \in A_4$ satisfies

$$(1\,2\,3)^{-1}(1\,2\,4)(1\,2\,3) = (1\,3\,2)(1\,2\,4)(1\,2\,3) = (1\,4\,3)$$

$(1\,2\,3)(1\,4\,3) = (1\,4)(2\,3)$, ... in this way, A_4 is generated by all 3-cycles in A_4, so $\ker\varphi = A_4$. By the first isomorphism theorem $A_4/\ker\varphi \cong \operatorname{Im}\varphi$ is trivial, so the action is trivial,
But, the left coset action is transitive, it cannot be trivial, Contradiction. $\square$

「Similar Problems.

1. Show that A_5 has no subgroup of order 15.

Proof. $|A_5| = |S_5|/2 = 60 = 2^2 \cdot 3 \cdot 5$.

Suppose that there is a subgroup $H < A_5$ of order 15.

By the Sylow Theorem, $\# Syl_3(H) \in \{1\}$ and $\# Syl_5(H) \in \{1\}$, that is, there are order 3 subgroup $P < H$ and order 5 subgroup $Q < H$, and both are normal subgroup of H. Since $P \cong C_3$ and $Q \cong C_5$ both are cyclic, let $x \in P$ and $y \in Q$ such that $P = \langle x \rangle$ and $Q = \langle y \rangle$.

Consider the afforded Homomorphism by H acts P by conjugation

$$\varphi : H \to S_P ; h \mapsto (k \mapsto hkh^{-1}).$$

Then, $\operatorname{Ker} \varphi = \{ h \in H ; hkh^{-1} = k \text{ for all } k \in P \} = C_H(P)$ and $H / \operatorname{Ker} \varphi \cong \operatorname{Im} \varphi \leq \operatorname{Aut}(P)$. But, since $\operatorname{Aut}(P) \cong \mathbb{Z}/2\mathbb{Z}$, and $|H|$ has no factor 2, odd number, so $|\operatorname{Ker} \varphi| = |H|$, that is, $C_H(P) = H$. Now, x and y commute, therefore $|xy| = 15$.

But, A_5 cannot have order 15 element, because:

order of permutation in S_5 is determined by the least common multiplier of disjoint cycle decompositions, but $\operatorname{lcm} 5 = 5$, $\operatorname{lcm}(4,1) = 4$, $\operatorname{lcm}(3,2) = 6$, $\operatorname{lcm}(2,1,1,1) = 2$, $\cdots$

the lcm cannot be 15, so it contradicts.

[Similar problem

Let $G = S_3 \times C_2$.

1. Find all normal subgroup of G.
2. Prove that G has no normal subgroup of order 4.

Proof. Define three maps:

$$\varphi_1 : G \to S_3 : (\sigma, a) \mapsto \sigma$$

$$\varphi_2 : G \to C_2 : (\sigma, a) \mapsto a$$

$$\varphi_3 : G \to \{\pm 1\} : (\sigma, a) \mapsto \operatorname{Sgn}\sigma \cdot \pi(a) \quad \text{where } C_2 \overset{\pi}{\cong} \{\pm 1\}.$$

$$\varphi_4 : G \to \{\pm 1\} \times C_2 : (\sigma, a) \mapsto (\operatorname{Sgn}\sigma, a).$$

$$\varphi_5 : G \to \{\pm 1\} : (\sigma, a) \mapsto \operatorname{Sgn}\sigma$$

Then, φ_1 and φ_2 are Homomorphisms, being the Coordinatively Projection, and φ_3 is also Homomorphism, because Sgn, π are Homomorphism and $\{\pm 1\}$ is abelian. Precisely,

$$\varphi_3((\sigma_1, a_1)\cdot(\sigma_2, a_2)) = \varphi_3(\sigma_1\sigma_2, a_1 a_2) = \operatorname{Sgn}\sigma_1 \operatorname{Sgn}\sigma_2\, \pi(a_1)\pi(a_2)$$
$$= \operatorname{Sgn}\sigma_1\, \pi(a_1) \operatorname{Sgn}\sigma_2\, \pi(a_2) = \varphi_3(\sigma_1, a_1)\cdot\varphi_3(\sigma_2, a_2).$$

And, φ_4 is Homomorphism, being entrywise Homomorphism, and φ_5 is Homomorphism, being Composition of Projection and Sgn.

Now, the Kernel

$\operatorname{Ker}\varphi_1 = \{id\} \times C_2$, $\operatorname{Ker}\varphi_2 = S_3 \times \{x^0\}$, $\operatorname{Ker}\varphi_3 = A_3 \times \{x^0\} \cup (S_3 \setminus A_3) \times \{x^1\}$

$\operatorname{Ker}\varphi_4 = A_3 \times \{x^0\}$, $\operatorname{Ker}\varphi_5 = A_3 \times C_2$.

are we desired.

Are these all possibilities?

Observe that; if $N \subseteq G_1 \times G_2$ is normal, then the image of projections $\pi_i : G_1 \times G_2 \to G_i : (g_1, g_2) \mapsto g_i$, $\pi_i[N]$ is also normal of $\pi_i[G_1 \times G_2] = G_i$, because image of normal subgroup is normal in the range, and the projections are onto.

But, as we know, normal subgroup of S_3 is only $\{id\}$, A_3, S_3 and normal subgroup of $\{\pm 1\}$ is $\{1\}$ and $\{\pm 1\}$.

[2018. Fall.

7. Consider the identity $\dfrac{e^{-x}}{1-x} = \sum_{n=0}^{\infty} C_n x^n$.

a). Determine the sequence $(C_n) \subseteq \mathbb{R}$ and the range of $x \in \mathbb{R}$ for the identity hold.

b). Show that for each $n \geq 0$,

$$\sum_{k=0}^{n} \frac{C_k}{(n-k)!} = 1.$$

Proof. a).

Notice that $e^{-x} = \sum_{n=0}^{\infty} \dfrac{(-x)^n}{n!}$ for all $x \in \mathbb{R}$ and

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad \text{for all } |x|<1.$$

Therefore, the coefficient C_n is determined by the Cauchy Product

$$C_n = \sum_{k=0}^{n} \frac{(-1)^{n-k}}{(n-k)!} \cdot 1^k = \sum_{k=0}^{n} \frac{(-1)^{n-k}}{(n-k)!}.$$

And, the identity valid in $\mathbb{R} \cap (-1,1) = (-1,1)$.] 5m.

b). Since

$$\frac{1}{1-x} = e^x \cdot \sum_{n=0}^{\infty} C_n x^n,$$ So we can write for $x \in (-1,1)$

$$\sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} \sum_{k=0}^{n} \frac{C_k}{(n-k)!} x^n.$$

And, the coefficients equals for each $n \geq 0$, because these are infinitely differentiable, So the coefficients of Powe series are determined uniquely, by the taylor Theorem. Thus,

$$1 = \sum_{k=0}^{n} \frac{C_k}{(n-k)!} \quad \text{for all } n \geq 0.$$] 4m.

[2019 Spring]

3. Rationalize the denominator of $\dfrac{1}{3+2\cdot\sqrt[3]{2}+\sqrt[3]{4}}$

Proof. Denote that $\alpha=\sqrt[3]{2}$. Then, $\alpha^3-2=0$.

Meanwhile, we want to find that the $\mathbb{Q}$-linear combination for

$$(3+2\cdot\sqrt[3]{2}+\sqrt[3]{4})^{-1}=(\alpha^2+2\alpha+3)^{-1}.$$

By the division algorithm,

$$\alpha^3-2=(\alpha^2+2\alpha+3)\cdot(\alpha-2)+\alpha+4$$

$$\alpha^2+2\alpha+3=(\alpha+4)(\alpha-2)+11.$$

$$\Rightarrow(\alpha^2+2\alpha+3)+(\alpha^2+2\alpha+3)(\alpha-2)^2=11 \qquad (\alpha^3-2=0)$$

$$\Rightarrow(\alpha^2+2\alpha+3)(\alpha^2-4\alpha+5)=11.$$

Therefore,

$$(\alpha^2+2\alpha+3)^{-1}=\frac{1}{11}\cdot(\alpha^2-4\alpha+5)=\frac{\sqrt[3]{4}-4\cdot\sqrt[3]{2}+5}{11}.$$

「2019, Spring

6. For each $n \in \mathbb{N}$, let
$$f_n(x) = \frac{|x|^n}{1+|x|^n}, \quad x \in \mathbb{R}.$$

a). Show that $(f_n)_{n\geq 1}$ does not converges uniformly.

b) For each fixed $q>1$, Show that (f_n) Converges uniformly on $\{x \in \mathbb{R} \mid |x| \geq q\}$ and $\{x \in \mathbb{R} \mid |x| \leq \frac{1}{q}\}$

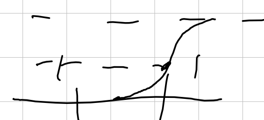
a). Given $x \in \mathbb{R}$,

if $|x|<1$, then $f_n(x) \to 0$ as $n \to \infty$, since $|x|^n \to 0$.

if $|x|=1$, then $f_n(x) = \frac{1}{2}$ for all $n \in \mathbb{N}$, since $|x|^n = 1$.

if $|x|>1$, then $f_n(x) = \dfrac{1}{1/|x|^n + 1} \to 1$ as $n \to \infty$, since $1/|x|^n \to 0$ as $n \to \infty$.

Therefore, $f_n \to f$ pointwise where
$$f(x) = \begin{cases} 0 & |x|<1 \\ \frac{1}{2} & |x|=1 \\ 1 & |x|>1. \end{cases}$$



But, for $\epsilon = 1/4$, and for all $n \in \mathbb{N}$,
$$\sup_{0\leq x<1} |f_n(x) - f(x)| = \sup_{0 \leq x<1} |f_n(x)| = \sup_{0\leq x<1} \frac{|x|^n}{1+|x|^n} \geq \frac{1}{4}$$
because given $n \in \mathbb{N}$, we can take $x \in [0,1)$ such that $\dfrac{x^n}{1+x^n} \geq \frac{1}{4}$

Since $\displaystyle\lim_{x\to 1^-} \frac{|x|^n}{1+|x|^n} = \frac{1}{2}$. So, $\displaystyle\sup_{x\in\mathbb{R}} |f_n - f| \geq \sup_{0\leq x<1} |f_n - f|$ implies $f_n \to f$ but not uniformly $\lrcorner$ lom.

b). If $|x| \geq q > 1$, then $f(x) = 1$. So, enough to show that $\sup |f_n(x) - f(x)| \to 0$.

Since $\displaystyle\sup_{|x|\geq q} |f_n(x) - f(x)| = \sup_{|x|\geq q} \left| \frac{1}{1+|x|^n} \right| \leq \frac{1}{1+|q|^n} \to 0$ as $n \to \infty$, so $f_n \to f$ uniformly on $|x| \geq q$. Similarly, for $|x| \leq \frac{1}{q}$, $f(x) = 0$ and so
$$\sup_{|x|\leq \frac{1}{q}} |f_n(x) - f(x)| = \sup_{|x|\leq\frac{1}{q}} \left| \frac{|x|^n}{1+|x|^n} \right| \leq \frac{1}{|q|^n + 1} \to 0 \text{ as } n \to \infty,$$

So Proved all cases. $\lrcorner$ lom. Q

「2019, Spring.

17. Let $S=\{(x,y,z)\in\mathbb{R}^3 \mid z=x\cdot f(\frac{y}{x}),\ x\neq 0\}$ where $f:\mathbb{R}\to\mathbb{R}$ is differentiable. Show that the tangent plane at any point of S passes through the origin.

Proof By the definition of S, given $(x_0,y_0,z_0)\in S$, a map

$X: \mathcal{O}\to S : (u,v)\mapsto (u,v,u\cdot f(\frac{v}{u}))$ where $\mathcal{O}$ is an open set containing (x_0,y_0).

is a parametrization of S at (x_0,y_0,z_0). The differential of X at (x_0,y_0) is:

$$dX=\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ f(\frac{v}{u})-\frac{v}{u}\cdot f'(\frac{v}{u}) & f'(\frac{v}{u}) \end{bmatrix}\Bigg|_{(u,v)=(x_0,y_0)}$$

So, the tangent plane of S at (x_0,y_0) is generated by

$$\left\{\begin{pmatrix} 1 \\ 0 \\ f(\frac{y_0}{x_0})-\frac{y_0}{x_0}\cdot f'(\frac{y_0}{x_0}) \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ f'(\frac{y_0}{x_0}) \end{pmatrix}\right\}.$$

Now, for $t_1,t_2\in\mathbb{R}$, the coordinate of a point of the tangent plane can be described by

$$(x_0,y_0,x_0\cdot f(\tfrac{y_0}{x_0}))+(t_1,0,t_1 f(\tfrac{y_0}{x_0})-t_1\cdot\tfrac{y_0}{x_0}f'(\tfrac{y_0}{x_0}))$$
$$+(0,t_2,t_2\cdot f'(\tfrac{y_0}{x_0}))$$

is zero when $t_1=-x_0$ and $t_2=-y_0$, so proved. ∎

2019. Fall.

1. Classify the following groups according to the FTFGA:

$$\frac{\mathbb{Z}\times\mathbb{Z}}{\langle(n,n)\rangle}$$

Proof. Define a map for each $n\in\mathbb{N}$:

$$\varphi : \mathbb{Z}\times\mathbb{Z} \to \mathbb{Z}_n\times\mathbb{Z} \; ; (a,b) \mapsto (a \bmod n, a-b).$$

Since

$$\begin{aligned}\varphi((a_1,b_1)+(a_2,b_2)) &= \varphi(a_1+a_2, b_1+b_2)\\ &= (a_1+a_2 \bmod n, a_1+a_2-b_1-b_2)\\ &= (a_1 \bmod n, a_1-b_1)+(a_2 \bmod n, a_2-b_2)\\ &= \varphi(a_1,b_1)+\varphi(a_2,b_2)\end{aligned}$$

So φ is Homomorphism, and given $(c,d)\in\mathbb{Z}_n\times\mathbb{Z}$,
take $a=c$ and $b=c-d$, then

$$\varphi(a,b)=(a,a-b)=(c,d)$$

So φ is onto. And,

$(a,b)\in \ker\varphi \iff a\equiv 0 \bmod n$ and $a-b=0$

So, $\ker\varphi=\langle(n,n)\rangle$. Now, by the first isomorphism theorem

$$\mathbb{Z}\times\mathbb{Z}/\ker\varphi \cong \operatorname{Im}\varphi = \mathbb{Z}_n\times\mathbb{Z}.$$

「2019, Fall.

2. Let p be the prime number and $\alpha \in \mathbb{F}_p$. Show that $x^p + \alpha$ is reducible over $\mathbb{F}_p$.

Proof. If $\alpha = 0$, then x^p is clearly reducible. Assume that $\alpha \neq 0$.
Let $a \in \overline{\mathbb{F}_p}$ be a root of $x^p + \alpha = 0$.
If $a \in \mathbb{F}_p$, then $x^p + \alpha$ has a linear factor over $\mathbb{F}_p$, so reducible.
Assume $a \notin \mathbb{F}_p$.

[2019, Fall.

8. Let $(b_n)_{n\in\mathbb{N}}$ be a decreasing sequence of positive real numbers with $b_n \to 0$. Let $(a_n)_{n\in\mathbb{N}}$ be a sequence of complex numbers such that $\left(\left|\sum_{n=1}^{N} a_n\right|\right)_{N\in\mathbb{N}}$ is bounded sequence.

a). Show that $\sum_{n=1}^{\infty} a_n b_n$ converges.

b). Show $\sum_{n=2}^{\infty} \frac{\sin(nx)}{\log n}$ converges for all $x \in [-\pi, \pi]$.

Proof. Denote $A_n = \sum_{k=1}^{n} a_k$ and $A_0 = 0$. And let $M > 0$ be a bound for $\sum a_n$.

$$\sum_{n=1}^{N} a_n b_n = \sum_{n=1}^{N} [A_n - A_{n-1}] \cdot b_n = \sum_{n=1}^{N} A_n b_n - \sum_{n=1}^{N} A_{n-1} \cdot b_n \quad \left(\sum_{n=1}^{N} A_{n-1} b_n = \sum_{n=0}^{N-1} A_n b_{n+1}\right)$$

$$= \sum_{n=1}^{N} A_n b_n - \left(\sum_{n=1}^{N} A_n \cdot b_{n+1} + A_0 b_1 - A_N b_{N+1}\right)$$

$$= \sum_{n=1}^{N} A_n (b_n - b_{n+1}) + A_N b_{N+1}.$$

Therefore, for any $m > n \geq 1$,

$$\sum_{k=n}^{m} a_k b_k = \sum_{k=n}^{m} A_k (b_k - b_{k+1}) + A_m b_{m+1} - A_{n-1} b_n$$

Now, using the hypothesis,

$$\left|\sum_{k=n}^{m} a_k b_k\right| \leq \sum_{k=n}^{m} |A_k| \cdot |b_k - b_{k+1}| + |A_m| |b_{m+1}| + |A_{n-1}| \cdot |b_n|$$

$$\leq M \cdot \left(\sum_{k=n}^{m} |b_k - b_{k+1}| + |b_{m+1}| + |b_n|\right)$$

$$= M \cdot \left(\sum_{k=n}^{m} (b_k - b_{k+1}) + b_{m+1} + b_n\right)$$

$$= M \cdot 2 b_n \to 0 \text{ as } n \to \infty, \text{ so proved.}$$

b). Since $1/\log n$ is decreasing and positive real numbers, so by a), enough to show that $\left|\sum \sin(nx)\right|$ is bounded for all $x \in [-\pi, \pi]$.

observe that: $2\sin x \cdot \sin nx = \cos(nx - x) - \cos(nx + x)$

Therefore, $2\sin x \cdot \sum_{n=1}^{N} \sin(nx) = \sum_{n=1}^{N} \cos[(n-1)x] - \cos[(n+1)x] \leq |\cos 0| + |\cos x| + |\cos Nx| + |\cos(N+1)x|$
≤ 4

So for $x \in [-\pi, \pi] \setminus \{0, \pi, -\pi\}$, $\left|\sum_{n=2}^{N} \sin nx\right| \leq \frac{2}{|\sin x|}$ for any $N \in \mathbb{N}$, bounded.

for $x = 0$ or $\pm\pi$, then the sum is just zero. ∎

[2019, Fall,

9. Let $\varphi : [0,1] \to \mathbb{R}$ be a continuous map. Evaluate

$$\lim_{\varepsilon \to 0} \int_0^1 \left[\frac{\varphi(x)}{x - i\varepsilon} - \frac{\varphi(x)}{x + i\varepsilon} \right] dx.$$

Proof. Observe that

$$\frac{1}{x - i\varepsilon} - \frac{1}{x + i\varepsilon} = \frac{2i\varepsilon}{x^2 + \varepsilon^2}$$

「2020. Spring.

6. Find the general solution of the differential equation

$$y''-3y'+2y=\frac{1}{1+e^{-t}}.$$

Proof. Since the characteristic of $y''-3y'+2y=0$ is $r^2-3r+2=(r-1)(r-2)$, therefore the general solution of the homogeneous part is:

$$y_h(t)=C_1e^t+C_2e^{2t} \quad (C_1, C_2\in\mathbb{R}).$$

And, to find a solution of the given equation, we need to find the particular solution $y_p(t)$. First, calculate the Wronskian,

$$\begin{vmatrix} e^t & e^{2t}\\ e^t & 2e^{2t}\end{vmatrix} = 2e^{3t}-e^{3t}=e^{3t}.$$

Therefore, the particular solution is

$$y_p(x)=\int_{x_0}^{x} \begin{vmatrix} e^t & e^x \\ e^{2t} & e^{2x}\end{vmatrix}\cdot\frac{1}{1+e^{-t}}\cdot\frac{1}{e^{3t}}\,dt$$

$$=\int_{x_0}^{x} \frac{e^{t+2x}-e^{2t+x}}{e^{3t}+e^{2t}}\,dt$$

$$=e^{2x}\int_{x_0}^{x}\frac{1}{e^{2t}+e^t}\,dt - e^x\int_{x_0}^{x} \frac{1}{e^t+1}\,dt$$

Meanwhile, since $\int \frac{1}{e^t+1}dt=\int\frac{e^{-t}}{1+e^{-t}}dt=-\log(1+e^{-x})$

and $\int\frac{1}{e^{2t}+e^t}dt=\int\frac{e^{-t}}{e^t+1}dt=\int\frac{-u}{\frac{1}{u}+1}\frac{1}{u}du=\int\frac{-u}{1+u}du$

$$=\int -1+\frac{1}{1+u}\,du=-u+\log(1+u)=-e^{-x}+\log(1+e^{-x})$$

Finally, the general solution is:

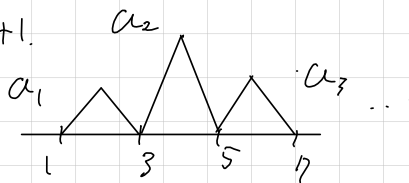
$$y_p+C_2\varphi_1+C_3\varphi_3=\left[-e^x+e^{2x}\log(1+e^{-x})+e^x\cdot\log(1+e^{-x})\right]$$

$$+C_2\cdot e^x+C_3e^{2x},\quad C_1,C_2,C_3\in\mathbb{R},$$

[2020 Spring.

7. Let $(a_n)_{n\in\mathbb{N}}$ be a sequence of non-negative real numbers and consider
$$f(x)=\sum_{n=1}^{\infty} a_n\cdot\max\{1-|x-2n|,0\},\quad x\in[0,\infty)$$
Give a necessary and sufficient condition on (a_n) such that
a). the series to converge uniformly on $[0,\infty)$.
b). f is continuous uniformly.

Proof. Notice that: given $x\geq 0$,
$$1-|x-2n|\geq 0 \iff 2n-1\leq x\leq 2n+1.$$
Therefore, f can be described as



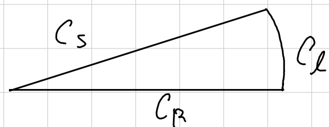
a). The given series converges uniformly if and only if
$$\sup_{x\geq 0}\left|f(x)-\sum_{n=1}^{N} a_n\cdot\max\{1-|x-2n|,0\}\right|\to 0 \quad \text{as } N\to\infty.$$
Meanwhile, the supremum term above equals to
$$\sup_{k\geq N+1} a_k$$
Therefore, $\lim_{n\to\infty}\sup a_n=0$ is the condition we desired.
(Since a_n is non-negative, $\lim_{n\to\infty}\sup a_n=\lim_{n\to\infty} a_n=0$)

b) If (a_n) is bounded by $M>0$, then, given $\varepsilon>0$, take $\delta=\frac{\varepsilon}{M}$. Then
$$|x-y|<\delta=\frac{\varepsilon}{M}\Rightarrow |f(x)-f(y)|\leq M\cdot|x-y|<\varepsilon,$$ So continuous uniformly.

Conversely, if (a_n) is unbounded, then
$$\forall\varepsilon>0,\ \exists\delta>0 \text{ such that } |x-y|<\delta\Rightarrow|f(x)-f(y)|<\varepsilon$$
is false, because for $\varepsilon=1$, for any $\delta>0$, there exists x_n, y_n such that $|f(x_n)-f(y_n)|=a_n\cdot|x_n-y_n|=a_n\cdot\frac{\delta}{2}>1$, we take $n\in\mathbb{N}$ such that $a_n>\frac{2}{\delta}$.
So, the condition that we desired is (a_n) is bounded. ∎
(Technically, we can take $x_n=2n$, $y_n=2n+\frac{\delta}{2}$.)

2020 Spring.

8. For each $n = 2, 3 \cdots$ evaluate
$$\int_0^\infty \frac{1}{1+x^n}\,dx.$$



Proof. Let $f(z) = \dfrac{1}{1+z^n}$. Take a Contour C as a Composition of :

$$C_R : t \mapsto t \quad (0 \le t \le R), \quad C_\ell : t \mapsto R \cdot e^{\frac{it}{n}} \quad (0 \le t \le 2\pi),$$
$$C_s : t \mapsto t \cdot e^{\frac{2\pi i}{n}} \quad (0 \le t \le R), \text{ that is, } C = C_R * C_\ell * (-C_s).$$

Inside C, $1+z^n$ has only one zero, $\exp(i\pi/n)$, so f has only one singularity inside C, the simple pole $\exp(i\pi/n)$, by the Residue theorem,
$$\int_C f = 2\pi i \cdot \mathrm{Res}(f, \exp(i\pi/n)) = \lim_{z \to \exp(i\pi/n)} (z - \exp(i\pi/n)) \cdot \frac{1}{1+z^n} \cdot 2\pi i$$
$$= \frac{2\pi i}{n \cdot \exp(i\pi/n)^{n-1}} = -\frac{\exp(i\pi/n)}{n} \cdot 2\pi i$$

Meanwhile,
$$\left| \int_{C_\ell} f \right| = \int_0^{2\pi} \left| \frac{1}{1+R^n \cdot e^{it}} \frac{Ri}{n} \cdot e^{\frac{it}{n}} \right| dt$$
$$= \int_0^{2\pi} \frac{R/n}{|1+R^n \cdot e^{it}|}\,dt \le \int_0^{2\pi} \frac{R/n}{|R^n - 1|}\,dt \to 0 \text{ as } R \to \infty.$$

and
$$\int_{C_s} f = \int_0^R \frac{e^{2\pi i/n}}{1+t^n e^{2\pi i}}\,dt = e^{\frac{2\pi i}{n}} \cdot \int_0^R \frac{1}{1+t^n}\,dt, \text{ So that}$$
$$-\frac{2\pi i}{n} \cdot \exp(i\pi/n) = \lim_{R \to \infty} \int_C f = \int_0^\infty \frac{1}{1+x^n}\,dx - \int_0^\infty \frac{1}{1+x^n}\,dx \cdot e^{\frac{2\pi i}{n}}$$
$$= \left(1 - e^{\frac{2\pi i}{n}}\right) \cdot \int_0^\infty \frac{1}{1+x^n}\,dx$$
$$\Rightarrow \int_0^\infty \frac{1}{1+x^n}\,dx = -\frac{2\pi i}{n} \cdot \exp(i\pi/n) \Big/ \left(1 - e^{\frac{2\pi i}{n}}\right).$$

$\blacksquare$

[2020. Spring

15. Let $r=r(\theta)$ be a curve in $\mathbb{R}^2$ defined in polar coordinates (r,θ) with $r>0$, $0\le\theta<2\pi$.

a). Express the arc length of the curve using $r(\theta), r'(\theta)$.

b). Express the curvature at $r(\theta)$ using $r(\theta), r'(\theta), r''(\theta)$.

Proof. Denote $x=r\cdot\cos\theta$ and $y=r\cdot\sin\theta$; $\alpha(\theta)=(x,y)$.

a). $\frac{dx}{d\theta}=r'\cos\theta-r\sin\theta$, $\frac{dy}{d\theta}=r'\sin\theta+r\cos\theta$.

therefore, $\int_0^{2\pi}\left|\frac{d\alpha}{d\theta}\right|d\theta=\int_0^{2\pi}\sqrt{r'^2+r^2}\,d\theta$.

b). Since the tangent vector with respect to the arc length s

$$t(\theta)=\frac{d\alpha}{ds}=\frac{1}{\frac{ds}{d\theta}}\cdot\frac{d\alpha}{d\theta}=\left|\frac{d\alpha}{d\theta}\right|^{-1}\cdot\frac{d\alpha}{d\theta}=\frac{1}{\sqrt{r'^2+r^2}}\cdot(r'\cos\theta-r\sin\theta,\ r'\sin\theta+r\cos\theta)$$

and since the α is plane curve, the normal vector is

$$n(\theta)=\frac{1}{\sqrt{r'^2+r^2}}\cdot(-r'\sin\theta-r\cos\theta,\ r'\cos\theta-r\cdot\sin\theta)$$

Since the curvature is

$$\kappa(\theta)=\left\langle\frac{dt}{ds},n\right\rangle=\left\langle\frac{d\theta}{ds}\cdot\frac{dt}{d\theta},n\right\rangle$$

$$=\left|\frac{d\theta}{d\alpha}\right|\cdot\frac{1}{(\sqrt{r'^2+r^2})^2}\cdot\Big\langle(r''\cos\theta-r'\sin\theta-r'\sin\theta-r\cos\theta,\ r''\sin\theta+r'\cos\theta+r'\cos\theta-r\sin\theta),$$
$$(-r'\sin\theta-r\cos\theta,\ r'\cos\theta-r\sin\theta)\Big\rangle$$

$$=\frac{1}{(r'^2+r^2)^{\frac{3}{2}}}\cdot\Big\langle((r''-r)\cos\theta-2r'\sin\theta,\ (r''-r)\sin\theta+2r'\cos\theta),$$
$$(-r'\sin\theta-r\cos\theta,\ r'\cos\theta-r\sin\theta)\Big\rangle$$

$$=\frac{1}{(r'^2+r^2)^{\frac{3}{2}}}\cdot(r''r+r^2+2r'^2)$$

Comment. 미분기하는 사라져야 한다.

$\boxed{\text{Debris}}$

observe that

$\frac{d\alpha}{d\theta}=\left(\frac{dx}{d\theta},\frac{dy}{d\theta}\right)$, so $t(\theta)=\frac{1}{\sqrt{r'^2+r^2}}\cdot(r'\cos\theta-r\cdot\sin\theta,\ r'\cdot\sin\theta+r\cos\theta)$

$n(\theta)=\frac{1}{\sqrt{r'^2+r^2}}\cdot(-r'\sin\theta-r\cdot\cos\theta,\ r'\cdot\cos\theta-r\cdot\sin\theta)$

$$t'(\theta)=\left(-\frac{1}{2}\cdot\frac{2r'r''+2rr'}{\sqrt{r'^2+r^2}}\right)\cdot\sqrt{r'^2+r^2}\cdot t(\theta)$$
$$+\frac{1}{\sqrt{r'^2+r^2}}\cdot(r''\cos\theta-r'\sin\theta-r'\sin\theta-r\cos\theta,\ r''\sin\theta+r'\cos\theta+r'\cos\theta-r\sin\theta)$$

Now, the curvature

$$\kappa(\theta)=\langle t',n\rangle=\frac{1}{(r'^2+r^2)^{\frac{3}{2}}}\Big\langle((r''-r)\cos\theta-2r'\sin\theta,\ (r''-r)\sin\theta+2r'\cos\theta),$$
$$(-r'\sin\theta-r\cos\theta,\ r'\cos\theta-r\sin\theta)\Big\rangle$$

$$=\frac{1}{(r'^2+r^2)^{\frac{3}{2}}}\cdot(r^2-rr''+2r'^2)$$

[2020, Fall.

1. Let p be a prime. Find the number of subgroups H of $\mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3}$, $|H| = p^2$

Proof. By the fundamental theorem of finitely generated abelian group, either $H \cong \mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \times \mathbb{Z}_p$.

If $H \cong \mathbb{Z}_p \times \mathbb{Z}_p$, then for all $x \in H$, $p \cdot x = 0$. Therefore, $H \subseteq \{(a,b) \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3} \mid p \cdot (a,b) = 0\}$.

Meanwhile, $pa = 0$ and $pb = 0$ if and only if $a \in \langle p \rangle \subseteq \mathbb{Z}_{p^2}$ and $b \in \langle p^2 \rangle \subseteq \mathbb{Z}_{p^3}$, so $\{(a,b) \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3} \mid p \cdot (a,b) = 0\} = \langle p \rangle \times \langle p^2 \rangle$.

Since $\langle p \rangle \times \langle p^2 \rangle$ has already order p^2, so $H = \langle p \rangle \times \langle p^2 \rangle$.

If $H \cong \mathbb{Z}_{p^2}$, that is, H is generated by order p^2 element,

Since $x \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3}$ has order $p^2 \iff p^2 \cdot x = 0$ and $px \neq 0$.

Every element $a \in \mathbb{Z}_{p^2}$ satisfies $p^2 a = 0$, and $b \in \mathbb{Z}_{p^3}$ satisfies $p^2 \cdot b = 0$ if and only if $b \in p\mathbb{Z}_{p^3}$ and $\mathbb{Z}_{p^3}$ has number of p^2 of p-multiples, so there are $p^2 \cdot p^2 = p^4$ elements satisfying $p^2 \cdot x = 0$.

And, $px = 0$ if and only if $a \in p\mathbb{Z}_{p^2}$ and $b \in p^2 \mathbb{Z}_{p^3}$, where $x = (a,b)$.

So, there are p^2 elements satisfying $px = 0$.

Consequently, there are $x^4 - x^2$ elements x such that $H = \langle x \rangle$.

Moreover, the cyclic group of order p^2 has $\varphi(p^2) = p^2 - p$ generators,

So the possible $H \cong \mathbb{Z}_{p^2}$ are

$$(p^4 - p^2)/(p^2 - p)$$

numbers.

Finally, the number of H is $1 + (p^4 - p^2)/(p^2 - p)$ ☒

Note: for a prime p, $\varphi(p^n) = p^n - p^{n-1}$.

「2020 Fall.

2. For each $n \in \mathbb{N}$, let $a_n = \cos(2\pi/n)$.

a). Prove that a_n is algebraic over $\mathbb{Q}$

b). Calculate $\deg_{\mathbb{Q}}(a_5)$.

Proof. a). Let $\xi = \exp(2\pi i/n)$. Then, $\xi^{-1} = \exp(-2\pi i/n) = \cos(2\pi/n) - i\sin(2\pi/n)$ gives that $a_n = \cos(2\pi/n) = (\xi + \xi^{-1})/2$.

Meanwhile, since $\xi^n = \exp 2\pi i = 1$, so $\xi^n - 1$ gives ξ is algebraic over $\mathbb{Q}$.

Therefore, ξ^{-1} is also algebraic, thus $a_n = \dfrac{\xi + \xi^{-1}}{2}$ is algebraic over $\mathbb{Q}$.

b). Let $w = \exp(2\pi i/5)$. Since w is the primitive 5th root of unity, $w^4 + w^3 + w^2 + w + 1 = 0$. And, dividing by w^2 then $2a_5 = w + w^{-1}$ implies

$$w^4 + w^3 + w^2 + w + 1 = 0 \Rightarrow w^2 + w^{-2} + w + w^{-1} + 1 = 0$$
$$\Rightarrow 4a_5^2 - 2 + 2a_5 + 1 = 0$$
$$\Rightarrow 4a_5^2 + 2a_5 - 1 = 0.$$

Since $4x^2 + 2x - 1$ is irreducible over $\mathbb{Q}$ by the discriminant $2^2 + 16 = 20$ but $\sqrt{20} \notin \mathbb{Q}$, so $\deg_{\mathbb{Q}}(a_5) = 2$. ☒

「Fail.

a) Notice that; De Moivre's formular gives

$$e^{inz} = (\cos nz + i \cdot \sin nz) = (\cos z + i \cdot \sin z)^n$$

Therefore

$$\cos n \cdot \frac{2\pi}{n} = 1 = \mathrm{Re}\left(\cos\frac{2\pi}{n} + i \cdot \sin\frac{2\pi}{n}\right)^n$$
$$= \mathrm{Re}\left(\sum_{k=0}^{n} \left(\cos\frac{2\pi}{n}\right)^k \cdot \left(i \cdot \sin\frac{2\pi}{n}\right)^{n-k} \cdot \binom{n}{k}\right)$$

Meanwhile, $\sin\left(\frac{2\pi}{n}\right)^2 = 1 - \cos\left(\frac{2\pi}{n}\right)^2$ and the real part of $\sum_{k=0}^{n} \binom{n}{k} \cdot \left(\cos\frac{2\pi}{n}\right)^k \cdot \left(i \cdot \sin\frac{2\pi}{n}\right)^{n-k}$ yield only $n-k$ is even, since i^{n-k} is real $\Leftrightarrow n-k$, so $\mathrm{Re}\left(\sum_{k=0}^{n} \binom{n}{k} \cdot \left(\cos\frac{2\pi}{n}\right)^k \cdot \left(i \cdot \sin\frac{2\pi}{n}\right)^{n-k}\right) - 1 = 0$ is evaluated $\cos\left(\frac{2\pi}{n}\right)$ for some polynomial over $\mathbb{Q}$, so a_n is algebraic over $\mathbb{Q}$.

b). From above argument,

$$1 = \cancel{\underset{k=0}{1 \cdot a_5^0 \cdot 0}} + \underset{k=1}{5 \cdot a_5^1 \cdot (1 - a_5^2)} - \underset{k=3}{10 \cdot a_5^3 \cdot (1 - a_5^2)^2} + \underset{k=5}{a_5^5}$$
$$= 5 \cdot a_5 - 5 \cdot a_5^3 - 10 a_5^3 \cdot (1 - 2a_5^2 + a_5^4) + a_5^5$$
$$= 5 \cdot a_5 - 15 a_5^3 + 11 a_5^5 - 10 \cdot a_5^7.$$

So, $10 a_5^7 - 11 a_5^5 + 15 a_5^3 - 5 a_5 + 1 = 0$.

[2020. Fall

3. Let $\mathbb{K}$ be a field. Show that $(x^3-y^5) \subseteq \mathbb{K}[x,y]$ is prime ideal.

Proof. Since $\mathbb{K}[x,y]$ is U.F.D., every irreducible element is prime.
Thus, enough to show that x^3-y^5 is irreducible.
Notice that: $\mathbb{K}[x,y] = \mathbb{K}[y][x]$ by definition, So $x^3-y^5 \in \mathbb{K}[y][x]$.
Since $\gcd(1, y^5) = 1$, x^3-y^5 is primitive, by the Gauss Lemma,
x^3-y^5 is irreducible over $\mathbb{K}[y] \iff x^3-y^5$ is irreducible over $\mathbb{K}(y)$.
But, since x^3-y^5 has degree 3, x^3-y^5 is reducible only if it has a linear factor, in other say, x^3-y^5 has a root in $\mathbb{K}(y)$.
Suppose that for some $f(y)/g(y) \in \mathbb{K}(y)$,

$$\frac{f(y)^3}{g(y)^3} - y^5 = 0.$$

Then, $f(y)^3 = y^5 \cdot g(y)^3$. But $\deg f(y)^3 \equiv 0 \mod 3$, and
$\deg y^5 \cdot g(y)^3 = 5 + \deg g(y)^3 \equiv 2 \mod 3$.
Therefore, there is no such a root, so x^3-y^5 is irreducible over $\mathbb{K}[y]$,
So it is irreducible over $\mathbb{K}[x,y]$.

Note:

$$K[x,y] = K[y][x] \xrightarrow{\text{Gauss}} K(y)[x] \xrightarrow{\text{irr. Test.}}$$

[Modified]
Prove that $\gcd(m,n)=1$ implies $x^m - y^n \in F[x,y]$ is irreducible.

「2020. Fall.

4. Let $W \subseteq V = \mathbb{R}^n$ be a subspace. Show that there exists a homogeneous system of linear equation whose solution space is W.

Proof. Let $\mathcal{B} = \{\alpha_1, \cdots, \alpha_k\}$ be a basis for W.

Then, we can extend the basis $\mathcal{B}$ to $\overline{\mathcal{B}} = \mathcal{B} \cup \{\alpha_{k+1}, \cdots, \alpha_n\}$ for V.

Now, consider the map:

$$f_j : V \to \mathbb{R} ; \sum c_i \alpha_i \mapsto c_j$$

Then, $W = \ker f_{k+1} \cap \cdots \cap \ker f_n$, so for the standard basis $\mathcal{S}$ for V,

$$A = \begin{pmatrix} [f_{k+1}]_{\mathcal{S}} \\ \vdots \\ [f_n]_{\mathcal{S}} \end{pmatrix}$$ satisfies $\ker A = W$, so ($[f]_{\mathcal{S}}$ is row vector representation).

$$A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$ is the system we desired.

「2021. Spring.

1. Show that no group of 2020 is simple.

Proof $2020 = 2^2 \cdot 5 \cdot 101$. Denote n_p as the number of Sylow p-subgroups.
By the Sylow theorem, $n_{101} \equiv 1 \bmod 101$ and $n_{101} \mid 20$, so n_{101} must be 1.
Therefore, the Sylow 101-subgroup is normal so proved.

「Without using Sylow theorem.
By the Cauchy's theorem, G has cyclic subgroup $H \leq G$ of order 101.
Denote $A = \{gHg^{-1} \mid g \in G\}$. Then, A is orbit of H from the action

$$G \times \{S \leq G\} \to \{S \leq G\} : (g, S) \mapsto gSg^{-1}.$$

Since the stabilizer of this action of H is $G_H = N_G(H)$, we obtain

$$\# A = |G : N_G(H)|.$$

And since $H \leq N_G(H)$,

$$|G : N_G(H)| \leq |G : H| = 2020/101 = 20, \text{ so } \# A \leq 20.$$

Now, consider the action

$$H \times A \to A : (h, K) \mapsto hKh^{-1}.$$

Given $K \in A$, the H-orbit of this action is $\mathrm{Orb}_H(K) = \{hKh^{-1} \mid h \in H\}$ and

$$\# \mathrm{Orb}_H(K) = |H : \mathrm{Stab}_H(K)|, \ \mathrm{Stab}_H(K) \leq H \Rightarrow |\mathrm{Stab}_H(K)| \in \{1, 101\}.$$

So, $\# \mathrm{Orb}_H(K) \in \{1, 101\}$ and $\mathrm{Orb}_H(K) \leq A$, so $\# \mathrm{Orb}_H(K) = 1$.
therefore, every element in H normalizes $K \in A$, that is, $H \leq N_G(K)$.
Now, for some $K \in A$, if $H \neq K$ then $HK \leq G$ and

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|} = 101^2$$

but, since $101^2 \nmid 2020$, so $A = \{H\}$. That is, $H \trianglelefteq G$. ⊠

$\sqrt{2}$021, spring.

3. Let p be a prime number, and L be the Galois extension $\mathbb{Q}(\xi)$ where ξ is a primitive p^2th root of unity. 55.

a). Show that all $\mathbb{Q}$-automorphisms of L are of the form

$$\xi \mapsto \xi^k, \quad p \nmid k, \ k \in \mathbb{Z}.$$

b). Show that L contains a subfield whose Galois group is $\mathbb{Z}/p\mathbb{Z}$.

Proof.

a). By definition, $\{\xi^0, \xi^1, \cdots, \xi^{p^2-1}\}$ is a set of roots of the polynomial

$$x^{p^2}-1=(x-1)(x^{p^2-1}+x^{p^2-2}+\cdots+x+1)$$

irreducible

Meanwhile, $x^{p^2-1}+x^{p^2-2}+\cdots+x+1$ has a factor $\prod_{\substack{1\le a\le p^2 \\ \gcd(a,p^2)=1}}(x-\xi^a)=P(x)$

So, since the automorphisms maps a root to the conjugate, so every automorphism of $L/\mathbb{Q}$ maps $\xi \mapsto \xi^k$, $\gcd(k,p^2)=1$, so, $p \nmid k$. Moreover, since $\mathbb{Q}(\xi)$ is a finite dimensional vector space generated by ξ^k $(k=0,1,\cdots)$, so if $\xi \mapsto \xi^k$ chosen, then the automorphism is determined. it proves a).

b). Sine $\varphi(p^2)=p^2-p$, where φ is the Euler-Phi function, the Galois extension of $P(x)$ over $\mathbb{Q}$ is contained in $\mathbb{Q}(\xi)$ and it has the degree of p^2-p, so, this extension K satisfies

$$|Gal(K/\mathbb{Q})|=p^2-p.$$ And, by the Cauchy's theorem,

it has a subgroup of order p, $G \le Gal(K/\mathbb{Q})$. Now, the fixed field $\mathbb{Q}(\xi)^G$ is we desired.

Revised in the Next

「2021. spring

17. Let f be a polynomial given by

$$f(z) = a_0 + a_1 z + \cdots + a_d z^d \quad \text{over } \mathbb{C},\ a_d \neq 0.$$

$\boxed{8m.}$

Evaluate the value $\displaystyle\lim_{R\to\infty} \int_{|z|=R} \frac{z \cdot f'(z)}{f(z)}\, dz$

Proof. By the fundamental theorem of Algebra, f has d-roots $z_1, \cdots, z_d \in \mathbb{C}$.

That is, $f(z) = a_d (z - z_1) \cdots (z - z_d)$

Moreover, $\dfrac{f'(z)}{f(z)} = \dfrac{1}{z - z_1} + \cdots + \dfrac{1}{z - z_d}$, just calculation.

Therefore, for $R > \max\{|z_1|, \cdots, |z_d|\}$, by the residue theorem,

$$\frac{1}{2\pi i} \int_{|z|=R} \frac{z \cdot f'(z)}{f(z)}\, dz = \sum_{n=1}^{d} \operatorname{Res}\left(\frac{z}{z - z_n}, z_n \right) = \sum_{n=1}^{d} \lim_{z \to z_n} z = z_1 + \cdots + z_n$$

Consequently, the value is

$$2\pi i \cdot (z_1 + \cdots + z_n) = 2\pi i \cdot \left(\frac{-a_{d-1}}{a_d} \right),$$

☒

[2022. Spring

1. Let G be a group of order 2021.

a). Prove that G is not simple.

b) Prove that G is cyclic

proof. Notice that: $2021 = 43 \cdot 47$.

a). By the Sylow theorem, the number n_{43} of Sylow 43-subgroup satisfies $n_{43} \equiv 1 \mod 43$ and $n_{43} \mid 47$, so $n_{43} = 1$.
Therefore, $Syl_{43}(G) = \{P\}$, and $gPg^{-1} \in Syl_{43}(P)$ for all $g \in G$ implies $gPg^{-1} = P$, so P is normal. Thus G is not simple.

b). Similarly, $n_{47} = 1$, so there is a unique Sylow 47-subgroup Q.
By the Lagrange's theorem, all non-zero elements in P and Q have order of 43 and 47, respectively, so $P \cap Q = \{e\}$.
Since both 43 and 47 are prime, P and Q are cyclic, denote $P = \langle x \rangle$ and $Q = \langle y \rangle$. Since $[x,y] = x^{-1}y^{-1}xy \in P \cap Q$, because P and Q both are normal, so $x^{-1}y^{-1}x \in Q$ and $y^{-1}xy \in P$, so $[x,y] = e$. Thus, x and y commute. Now, xy is order of $43 \cdot 47$, so $G = \langle xy \rangle$ is cyclic.

[2022 Spring

2. Prove or disprove that each of following is PID:

a) $\mathbb{R}[x]$

b) $\mathbb{R}[x,y]$

c) $\mathbb{Z}[\sqrt{-1}]$

d) $\mathbb{Z}[x]$

Proof.

a) $\mathbb{R}[x]$ is PID, because since $\mathbb{R}$ is a field, so $\mathbb{R}[x]$ is Euclidean domain, so for any ideal $I \subseteq \mathbb{R}[x]$, we can find a generator of I

b) $\mathbb{R}[x,y]$ is not P.I.D., because: let $(x,y) \subset \mathbb{R}[x,y]$.
If $(x,y) = (P(x,y))$ for some $P \in \mathbb{R}[x,y]$,
then $P(0,0) = 0$, that is, P has zero constant term.
But, $x, y \in (x,y) = (P(x,y))$ implies $\deg_x P$, $\deg_y P \leq 1$,

「2022. Spring.

3. Find the Galois group of the polynomial $x^5+3x^3-2x^2-6$ over $\mathbb{Q}$.

Proof. Notice that

$$x^5+3x^3-2x^2-6=x^2\cdot(x^3-2)+3(x^3-2)=(x^3-2)\cdot(x^2+3).$$

Since the irreducible factor x^3-2 has three roots $\sqrt[3]{2}$, $\sqrt[3]{2}\cdot e^{\frac{2\pi}{3}i}$, $\sqrt[3]{2}\cdot e^{\frac{4\pi i}{3}}$ and the irreducible factor x^2+3 has two roots $\pm\sqrt{-3}$,

The splitting field is

$$\mathbb{K}=\mathbb{Q}\left(\sqrt[3]{2},\sqrt[3]{2}\cdot e^{\frac{2\pi i}{3}},\sqrt[3]{2}\cdot e^{\frac{4\pi i}{3}},\sqrt{-3},-\sqrt{-3}\right)=\mathbb{Q}(\sqrt[3]{2},e^{\frac{2\pi i}{3}},\sqrt{-3})$$

Moreover, since $\mathbb{Q}$ is perfect, so $\mathbb{K}/\mathbb{Q}$ is Galois.

Meanwhile, notice that

$$e^{\frac{2\pi i}{3}}=-\frac{1}{2}+\frac{\sqrt{-3}}{2}$$

Therefore, $\mathbb{K}=\mathbb{Q}(\sqrt[3]{2},e^{\frac{2\pi i}{3}})$. Finally,

$$[\mathbb{Q}(\sqrt[3]{2}):\mathbb{Q}]=3$$

and since $w=e^{\frac{2\pi i}{3}}$ satisfies w^2+w+1, so

$$[\mathbb{Q}(\sqrt[3]{2},w):\mathbb{Q}(\sqrt[3]{2})]\leq 2$$

But since $w\in\mathbb{C}\setminus\mathbb{R}$ and $\mathbb{Q}(\sqrt[3]{2})\subseteq\mathbb{R}$, so the degree is actually 2.

Since $Gal(\mathbb{K}/\mathbb{Q})\subseteq Sym(\{\sqrt[3]{2},\sqrt[3]{2}w,\sqrt[3]{2}w^2\})$, the tower lemma gives $[\mathbb{Q}(\sqrt[3]{2},w):\mathbb{Q}]=6$, so $|Gal(\mathbb{K}/\mathbb{Q})|=6$, that is,

$$Gal(\mathbb{K}/\mathbb{Q})\cong S_3.$$

「2022. Spring.

8. Define
$$f:\mathbb{R}\to\mathbb{R};\ x\mapsto\sum_{n=1}^{\infty}\frac{1}{x^2+n^2}$$
Then, f is differentiable on $\mathbb{R}$.

proof. For each $m\in\mathbb{N}$, define
$$f_m(x)=\sum_{n=1}^{m}\frac{1}{x^2+n^2}.$$
At first, $f_m(0)\to f(0)$ as $m\to\infty$, because that
$$\sum_{n=2}^{m}\frac{1}{n^2}<\sum_{n=2}^{m}\frac{1}{n^2-n}=\sum_{n=2}^{m}\frac{1}{n-1}-\frac{1}{n}=1-\frac{1}{m}\to 1 \text{ as } m\to\infty,$$
so the increasing bounded sequence $f_m(0)$ converges. Now,
$$f_m'(x)=\sum_{n=1}^{m}-\frac{2x}{(x^2+n^2)^2}$$
converges uniformly, because

if $|x|\le 1$, then
$$\left|\frac{-2x}{(x^2+n^2)^2}\right|\le\frac{2}{n^4}\quad\text{and}\quad\sum\frac{1}{n^4}<\infty,$$
if $|x|>1$, then
$$\left|\frac{-2x}{(x^2+n^2)^2}\right|\le 2\cdot\left|\frac{x}{x^2+n^2}\cdot\frac{1}{x^2+n^2}\right|\le 2\cdot\frac{1}{n^2}\quad\text{and}\quad\sum\frac{1}{n^2}<\infty$$
therefore, the Weierstrass M-test gives f_m' converges uniformly.

Now, f_m converges at some point and f_m' converges uniformly on $\mathbb{R}$, the limit function f is differentiable.」 $\blacksquare$

First, for any $x\in\mathbb{R}$, $|x^2+n^2|\ge n^2$, so
$$\sum_{n=1}^{\infty}\left|\frac{1}{x^2+n^2}\right|\le\sum_{n=1}^{\infty}\frac{1}{n^2}<\infty,\ \text{because}\ \sum_{n=2}^{m}\frac{1}{n^2}<\sum_{n=2}^{m}\frac{1}{n^2-n}=\sum_{n=2}^{m}\frac{1}{n-1}-\frac{1}{n}=1-\frac{1}{m}\to 1,$$
as $m\to\infty$. Therefore, given $\varepsilon>0$, for some $N\in\mathbb{N}$, $m>n\ge N$ implies
$$\sup_{x\in\mathbb{R}}\left|\sum_{k=n}^{m}\frac{1}{x^2+k^2}\right|\le\sup_{x\in\mathbb{R}}\sum_{k=n}^{m}\left|\frac{1}{x^2+k^2}\right|<\sum_{k=n}^{m}\frac{1}{k^2}<\varepsilon.$$
Now, define for each $m\in\mathbb{N}$,
$$f_m(x)=\sum_{n=1}^{m}\frac{1}{x^2+n^2}.$$
We already shown $f_m\to f$ uniformly. And,
$$f_m'(x)=\sum_{n=1}^{m}-\frac{2x}{(x^2+n^2)^2}$$
converges uniformly, because

「2022. spring.

11. Prove or disprove that:

a). A compact subset of any topological space is closed.

b). A connected metric space having more than one point is uncountable.

Proof.

a). Let $\mathbb{Z}_3 = \{0,1,2\}$ with indiscrete topology $\mathcal{T} = \{\mathbb{Z}_3, \emptyset\}$.
Then, a singletone $\{1\}$ is not closed, but it is compact set, because every open cover of $\mathbb{Z}_3$ is finite. So, false.

b) Let $a, b \in X$ be distinct elements of a connected metric space X
Given $\lambda \in (0,1)$, if there is no point $x \in X$ such that $d(a,x) = \lambda \cdot d(a,b)$,
then $\mathcal{U} = \{x \in X \mid d(a,x) < \lambda \cdot d(a,b)\}$ and $V = \{x \in X \mid d(a,x) > \lambda \cdot d(a,b)\}$
forms a separation of X, so it contradicts to the connecteedness.
($\mathcal{U} \cap V = \emptyset$ and $\mathcal{U}, V$ are open is clear, and
$\mathcal{U} \cup V = X$ is obtained by the assumption).
So, given $\lambda \in (0,1)$, there is a point $x_\lambda \in X$ such that $d(a, x_\lambda) = \lambda \cdot d(a,b)$.
Thus, if $\lambda, \lambda' \in (0,1)$ are different, then

$$d(a, x_\lambda) \neq d(a, x_{\lambda'})$$

So the map $(0,1) \to X : \lambda \mapsto x_\lambda$ is well-defined indective map,
thus X is uncountable. ☒

「2022. Fall.

1. Let G be a group and H and K are normal subgroup of G. If $H \cap K = \{e\}$, then $HK \cong H \times K$.

Proof. Define a map

$$\varphi : H \times K \to HK : (h, k) \mapsto hk.$$

It is well-defined by the definition of HK, and it is homomorphism; given $h_1, h_2 \in H$ and $k_1, k_2 \in K$,

$$\varphi((h_1, k_1) \cdot (h_2, k_2)) = \varphi((h_1 h_2, k_1 k_2)) = h_1 h_2 k_1 k_2$$

meanwhile, given $h \in H$ and $k \in K$, $hkh^{-1}k^{-1} \in K$ and $\in H$, because $hkh^{-1} \in K$ and $kh^{-1}k^{-1} \in H$, by the normality. But, $H \cap K = \{e\}$ implies $hkh^{-1}k^{-1} = e$, so $hk = kh$, these commute so,

$$h_1 h_2 k_1 k_2 = h_1 k_1 h_2 k_2 = \varphi((h_1, k_1))\, \varphi((h_2, k_2)),$$ so φ preserves the operation.

And, $\varphi[H \times K] = HK$ by just definition, φ is onto. Now, the first isomorphism theorem gives that $H \times K / \ker \varphi \cong HK$.

And, $$\ker \varphi = \{(h, k) \in H \times K \mid hk = e\} = \{e\},$$

so $H \times K \cong H \times K / \{e\}$, it proves the assertion. ∎

2022 Fall.

8. For continuous functions $g:[0,1]\to\mathbb{R}$ and $h:[-1,1]\to\mathbb{R}$, define

$$f:[0,1]\to\mathbb{R}: x\mapsto \int_0^1 h(x-y)g(y)\,dy$$

For each $n\in\mathbb{N}$, define

$$f_n:[0,1]\to\mathbb{R}: x\mapsto \frac{1}{n}\cdot\sum_{i=0}^{n-1} h\left(x-\frac{i}{n}\right)\cdot g\left(\frac{i}{n}\right)$$

Prove that $f_n\to f$ uniformly on $[0,1]$

Proof. Define $H:[0,1]\times[0,1]:(x,y)\mapsto h(x-y)\cdot g(y)$.

Since h and g both are continuous, therefore H is continuous and since $[0,1]^2$ is compact set, so H is continuous uniformly.

Given $\varepsilon>0$, there exists $\delta>0$ such that for any $(x_1,y_1),(x_2,y_2)\in[0,1]^2$,

$$\|(x_1,y_1)-(x_2,y_2)\|<\delta \Rightarrow \|H(x_1,y_1)-H(x_2,y_2)\|<\varepsilon.$$

Now, given $x\in[0,1]$, and $n>\frac{1}{\delta}$, ($n>\frac{1}{\delta}$ gives $\frac{1}{n}<\delta$)

$$\begin{aligned}
|f(x)-f_n(x)| &= \left|\int_0^1 H(x,y)\,dy - f_n(x)\right| \\
&= \left|\sum_{i=0}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}} H(x,y)\,dy - \frac{1}{n}\cdot\sum_{i=0}^{n-1} H\left(x,\frac{i}{n}\right)\right| \\
&= \left|\sum_{i=0}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}} H(x,y)\,dy - \sum_{i=1}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}} H\left(x,\frac{i}{n}\right)dy\right| \\
&\le \sum_{i=0}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}} \left|H(x,y)-H\left(x,\frac{i}{n}\right)\right| dy \\
&< \sum_{i=0}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}} \varepsilon\,dy = \varepsilon.
\end{aligned}$$

Therefore, $n>\frac{1}{\delta} \Rightarrow \sup_{x\in[0,1]}|f_n(x)-f(x)|<\varepsilon$, So, proved. ∎

2022. Fall.

12.

「2022. Fall.

13. Let S be the subset of $\mathbb{R}^3$ given by $\{(x,y,z) \mid z^2 = x^2+y^2,\ z \geq 0\}$.
Show that S is not regular surface.

proof. Suppose that S is a regular surface. Then, for $(0,0,0) \in S$,
there exists a neighbor V containing $(0,0,0)$ such that $V \cap S$ is a graph of

$$z = f(x,y) \quad \text{or} \quad y = g(x,z) \quad \text{or} \quad z = h(y,z)$$

where f, g, h are differentiable. Since projection of S into xz, yz-plain are not 1-1, g, h cannot be such a map.
Meanwhile, $f(x,y) = \sqrt{x^2+y^2}$ at locally $(0,0,0)$, but f is not differentiable at $(x,y) = (0,0)$, so it cannot be possible. Therefore S is not regular.

(More specifically, $(0, \pm\varepsilon, \sqrt{1-\varepsilon^2}) \in S$ for any $\varepsilon > 0$, so there is no 1-1 yz-projection, similarly xz.)

「2023 spring.

1). Let
$$A=\begin{bmatrix}0&-3&7\\-1&-1&5\\-1&-2&6\end{bmatrix}$$
a). Find the minimal polynomial of A.
b). 〃 Jordan form 〃
c). 〃 rational form 〃

proof.

※: tr = sum eigen.

a). First, the characteristic polynomial $\chi(t)$

$$\det\begin{bmatrix}t&3&-7\\1&t+1&-5\\1&2&t-6\end{bmatrix}=t\cdot[(t+1)(t-6)+10]-3\cdot[t-6+5]-7\cdot[2-1-t]$$
$$=t\cdot[t^2-5t+4]-3\cdot[t-1]+7[t-1]$$
$$=t\cdot[t-4]\cdot[t-1]+4[t-1]$$
$$=(t-1)(t-2)^2.$$

Since the minimal polynomial have same roots to χ, so we need to check that $(A-I)(A-2I)=0$. (If not, then Cayley-Hamilton theorem guarantees that χ is the minimal polynomial.)

$$(A-I)\cdot(A-2I)=\begin{bmatrix}-1&-3&7\\-1&-2&5\\-1&-2&5\end{bmatrix}\cdot\begin{bmatrix}-2&-3&7\\-1&-3&5\\-1&-2&4\end{bmatrix}\neq 0$$

So, $\chi=(t-1)(t\ 2)^2$.

b). $\dim\ker(A-I)=1$, since $0<\dim\ker(A-I)\leq\deg(t-1)=1$, and $\dim\ker(A-2I)\leq 2=\deg(t-2)^2$. Meanwhile,
$$\begin{bmatrix}2&-3&7\\-1&-3&5\\-1&-2&4\end{bmatrix}$$
has more than one linearly independent row vectors, so the nulity is 1. So, the Jordan form is

$$J_A=\begin{bmatrix}\boxed{1}&&\\&2&1\\&&2\end{bmatrix}$$

c). The minimal polynomial is equal to the characteristic, so A is cyclic, thus the Rational form is

$$R_A=\begin{bmatrix}0&0&4\\1&0&-8\\0&1&5\end{bmatrix}.$$

[2023. Spring

4. Let $V = \{id, (12)(34), (13)(24), (14)(23)\} \subseteq S_4$, 5?,

a). Show that V is normal subgroup.

b). $S_4/V \cong S_3$,

a). Since conjugate cannot change the order of elements, and cannot change the Parity; since the sign function is Homomorphism,

$$sgn(\sigma\tau\sigma^{-1}) = sgn(\sigma)\cdot sgn(\sigma^{-1})\cdot sgn(\tau) = sgn(\tau), \text{ for any } \sigma, \tau \in S_n.$$

Now, the fact that V contains all order two even permutations in S_4, so every conjugate of V is contained in V, so normal. 5m

b). Consider S_3 is a subgroup of S_4, the number 4 fixed. Then,

$$\varphi: S_3 \to S_4/V : \sigma \mapsto \sigma V$$

is well-defined, and given $\sigma_1, \sigma_2 \in S_3 \leq S_4$,

$$\varphi(\sigma_1\sigma_2) = \sigma_1\sigma_2 V = \sigma_1 V \sigma_2 V = \varphi(\sigma_1)\varphi(\sigma_2)$$

So it Homomorphism. And, since all elements in S_3 fixes 4, $S_3 \cap V = \{id\}$, so $\ker \varphi = \{id\}$. Now, since $|S_3| = 6 = |\ker\varphi|\cdot|Im\,\varphi|$, $|Im\,\varphi| = 6$, so $Im\,\varphi = S_4/V$, since $|S_4/V| = 6$, so φ is onto.

Now, the first isomorphism theorem gives $S_3 \cong S_4/V$. ☒

「2023 spring

5. Let $I=(2, x^2+x+1) \subseteq \mathbb{Z}[x]$ be an ideal.

Show that I is maximal ideal and give a multiplication table for $\mathbb{Z}[x]/I$.

proof. Define a map

$$\pi : \mathbb{Z}[x] \to \mathbb{Z}/2\mathbb{Z}[x] ; \sum_{n=0}^{n} a_n x^n \mapsto \sum_{n=0}^{n} (a_n \bmod 2) \cdot x^n$$

Then, π is an onto Homomorphism, ($\pi(0)=0$, $\pi(1)=1$) with $\operatorname{Ker}\pi = 2\mathbb{Z}[x] = (2)$, So $\mathbb{Z}[x]/\operatorname{Ker}\pi = \mathbb{Z}[x]/2\mathbb{Z}[x] \cong (\mathbb{Z}/2\mathbb{Z})[x]$.

Meanwhile, x^2+x+1 is irreducible over $\mathbb{Z}/2\mathbb{Z}$, because it is of degree 2 and $0^2+0+1=1\neq 0$, $1^2+1+1=1\neq 0$ in $\mathbb{Z}/2\mathbb{Z}$.

Now, Consider the natural Homomorphism

$$\varphi : \mathbb{Z}/2\mathbb{Z}[x] \to \left(\mathbb{Z}/2\mathbb{Z}[x]\right)/(x^2+x+1) ; p(x) \mapsto p(x) + (x^2+x+1)$$

It is trivially onto, So $\varphi\circ\pi$ is onto Homomorphism.

Moreover,

$$\operatorname{Ker}\varphi\circ\pi = (\varphi\circ\pi)^{-1}[\{0\}] = \pi^{-1}[\varphi^{-1}[\{0\}]] = \pi^{-1}[\operatorname{Ker}\varphi] = \pi^{-1}[(x^2+x+1)]$$

Now,

$$\begin{aligned} f\in\pi^{-1}[(x^2+x+1)] &\iff \pi(f)\in(x^2+x+1) \\ &\iff \pi(f) = (x^2+x+1)\cdot\pi(g) \quad \text{for some } g(x)\in\mathbb{Z}[x] \\ &\iff \pi(f-(x^2+x+1)g) = 0 \\ &\iff f-(x^2+x+1)g \in (2) \\ &\iff f\in(2)+(x^2+x+1) = (2, x^2+x+1). \end{aligned}$$

So, $\mathbb{Z}[x]/\operatorname{Ker}(\varphi\circ\pi) = \mathbb{Z}[x]/(2, x^2+x+1) \cong (\mathbb{Z}/2\mathbb{Z})[x]/(x^2+x+1)$.

And, Since $\mathbb{Z}/2\mathbb{Z}$ is field and x^2+x+1 is irreducible over $\mathbb{Z}/2\mathbb{Z}$, so $(\mathbb{Z}/2\mathbb{Z})[x]/(x^2+x+1)$ is a field, so, the quotient is field if and only if (x^2+x+1) is maximal, so proved.

Since $\mathbb{Z}[x]/I \cong (\mathbb{Z}/2\mathbb{Z})[x]/(x^2+x+1)$, Let $\alpha = x+(x^2+x+1)$.

Then $\alpha^2 = x^2+(x^2+x+1) = +x+1+(x^2+x+1) = +\alpha+1$, So

	1	α
1	1	α
α	α	$+\alpha+1$

is the table.

Generalized in the Next page.

「Let $I=(a,f)\subseteq R[x]$ be an ideal generated by $a, f\in R[x]$.

Consider the diagram:

$$\begin{tikzcd}
R[x] \arrow[r, "\varphi"] \arrow[dr, "\pi"'] & (R/aR)[x] \arrow[r, "\pi_f"] & (R/aR)[x]/(\bar{f}). \\
& R[x]/aR[x] \arrow[u, "\simeq"'] &
\end{tikzcd}$$

$R[x]\to R[x]/aR[x]$ $\qquad$ $R[x]/aR[x]\to (R/aR)[x]$

$: P(x)\mapsto P(x)+aR[x]$ $\qquad$ $: P(x)+aR[x]\mapsto \overline{P(x)}$, where $\overline{\sum a_n\cdot x^n}=\sum (a_n+aR)\cdot x^n$.

Onto Homomorphism. $\qquad$ Isomorphism.

$\Rightarrow$ $\varphi : R[x]\to (R/aR)[x]$ is Onto Homomorphism, So is $\pi_f\circ\varphi$ onto Homomorphism.

Moreover,

$$\begin{aligned}
\ker(\pi_f\circ\varphi) &= (\pi_f\circ\varphi)^{-1}[\{0\}] = \varphi^{-1}[\ker \pi_f] \\
&= \{P(x)\in R[x] \mid \overline{P(x)}\in \ker\pi_f\} = \{P(x)\in R[x] \mid \overline{P(x)}\in (\overline{f(x)})\} \\
&\overset{(*)}{=} \{P(x)\in aR[x]+f(x)R[x]\} = (a)+(f) = (a,f) = I.
\end{aligned}$$

Asumme $a\in R$ and $(a)\subseteq R$ is maximal, and $\bar{f}\in (R/aR)[x]$ is irreducible.

Then, R/aR is a field, and therefore $(R/aR)[x]$ is P.I.D.

So, $(\bar{f})$ is maximal, it implies that $(R/aR)[x]/(\bar{f})$ is a field.

Now, by the first-isomorphism theorem,

$$R[x]/(aR[x]+fR[x]) \cong (R/aR)[x]/\bar{f}\cdot(R/aR)[x]$$

Note: $(*)$ is obtained by: given $P(x)\in R[x]$,

$$\begin{aligned}
\overline{P(x)}\in (\overline{f(x)}) &\Leftrightarrow \overline{P(x)} = \overline{g(x)}\cdot\overline{f(x)} \text{ for some } g(x)\in R[x] \\
&\Leftrightarrow P(x)-g(x)f(x)\in aR[x] \text{ for some } g(x)\in R[x]
\end{aligned}$$

「Another Solution.

Observe that $\mathbb{Z}[x]/(2, x^2+x+1)$.

Given $f(x) \in \mathbb{Z}[x]$, Since the leading coefficient of x^2+x+1 is a unity, So we can apply the division Algorithm,

$$f(x) = g(x)\cdot(x^2+x+1) + r(x) \text{ for some } g, r \in \mathbb{Z}[x] \text{ with } \deg r \leq 1.$$

Therefore, $\mathbb{Z}[x]/(2, x^2+x+1) = \{ax+b + (2, x^2+x+1) \mid a, b \in \mathbb{Z}\}$

And, for any $a, b \in \mathbb{Z}$, $ax+b = (a \bmod 2)\cdot x + (b \bmod 2) + 2nx + 2m$

for some $n, m \in \mathbb{Z}$, so $= \{ax+b + (2, x^2+x+1) \mid a, b \in \{0, 1\}\}$.

And, $1 \notin (2, x^2+x+1)$ and $x \notin (2, x^2+x+1)$ and $x+1 \notin (2, x^2+x+1)$ implies these are distinct with zero, and also these are mutually distinct, just simple calculation. Let $\alpha = x + I$.

Now, $\mathbb{Z}[x]/I$ has four element, $\{0, 1, \alpha, \alpha+1\}$ where $\alpha^2+\alpha+1 = 0$, so the table

	0	1	α	$\alpha+1$
0	0	0	0	0
1	0	1	α	$\alpha+1$
α	0	α	$\alpha+1$	1
$\alpha+1$	0	$\alpha+1$	1	α

is obtained. Since every non-zero element has an inverse, it is a field,

Comment. Multiplication table 요구한 부분에서 이게 더 출제 의도에 가까운듯?

2023, Spring.

7. Let $f: \mathbb{C} \to \mathbb{C}$ be an entire satisfying 10m

$$u^2(z) - 2 \cdot v^2(z) = 1 \text{ for all } z \in \mathbb{C}$$

Prove that f is constant.

Proof. observe that ~~i differentiating gives~~ that

$$\cancel{2u u_x - 2 v v_x = 0 \Rightarrow u \cdot u_x = v v_x = -v \cdot u_y}$$

observe that: $f(z) = 0 \iff u(z) = v(z) = 0$

But, $u^2 - 2v^2 = 1$, so $f(z) \neq 0$ for all $z \in \mathbb{C}$. Now,

$$\frac{1}{f^2(z)} = \frac{1}{u^2(z) + v^2(z)} = \frac{1}{1 + 3v^2(z)} \leq 1,$$

therefore by the Liouville's theorem, the bounded analytic function $\frac{1}{f^2}$ is constant, so f is also constant function. ☑

Or, $|f^2| = u^2 + v^2 = 1 + 3v^2 \Rightarrow \frac{1}{f} \leq \frac{1}{\sqrt{1+3v^2}} \leq 1$.

2023 spring.

10. Let $f:[0,\infty)\to[0,\infty)$ be a continuous function satisfying

$$\int_0^\infty f(x)\,dx<\infty.$$

a). Prove or disprove that $\lim_{x\to\infty} f(x)=0$.

b). If f is continuous uniformly, then $\lim_{x\to\infty} f(x)=0$? Prove or disprove.

Proof.

a). Define a map on $[0,\infty)$

$$f_n(x)=\begin{cases}0 & \text{otherwise}\\ n^2\cdot(x-n) & \text{if } x\in[n,n+\frac{1}{n^2}]\\ n^2\cdot(n+\frac{2}{n^2}-x) & \text{if } x\in[n+\frac{1}{n^2},n+\frac{2}{n^2}]\end{cases}$$

Then $\int_0^\infty f_n=\frac{1}{n}$ and f_n is continuous, for each $n\in\mathbb{N}$.

Let $f=f_1+f_4+\cdots+f_{n^2}+\cdots$. Then, f is continuous and

$$\int_0^\infty f=\sum_{n=1}^\infty\frac{1}{n^2}<\infty.$$

(The continuity is obtained just investigate each $[n,n+\frac{2}{n^2}]$)

But, for $\epsilon=\frac{1}{2}$, for any $\delta>0$, there is $x>\delta$ such that $|f(x)-0|=1\geq\frac{1}{2}$.

So $\lim f(x)\neq0$.

b). Suppose $\lim_{x\to\infty} f(x)=0$ is false but f is continuous uniformly.

Let $\epsilon=2$. Then, for any $n\in\mathbb{N}$, there is $x_n>n$ such that $f(x_n)\geq2$.

Since f is uniformly continuous, take $1>\delta>0$ such that $|x-y|<\delta$ implies $|f(x)-f(y)|<1$. Then, for each $n\in\mathbb{N}$,

$$\int_{x_n}^{x_n+\delta} f(x)\,dx\geq\delta\cdot1$$

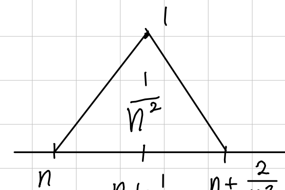
So, $\int_0^\infty f\geq\sum_{n=1}^N\int_{x_n}^{x_n+\delta} f\geq\delta\cdot N$ diverges, so contradicts.

Thus $\lim f=0$. ☒

Revised in the Next.

⌜Let $f:[0,\infty)\to\mathbb{R}$ be non-negative valued continuous function,
does not converges to zero,
but the improper integral converges.
Find and convince concretely, and change the condition as f continuous uniformly.

Define $f_n(x)=\begin{cases} n^2(x-n) & \text{if } x\in[n, n+\frac{1}{n^2}] \\ n^2(n+\frac{2}{n^2}-x) & \text{if } x\in[n+\frac{1}{n^2}, n+\frac{2}{n^2}] \\ 0 & \text{otherwise.}\end{cases}$



Then, f_n is continuous, with $\int_0^\infty f_n = \frac{1}{n^2}$.
Now, let $f=\sum_{n=2}^{\infty} f_n$. Since the interval $[n, n+\frac{2}{n^2}]$ are disjoint mutually, $n\geq 2$,
So the f is well-defined, but it does not uniformly converges, because

$$\sup_{x\geq 0}\left|f(x)-\sum_{n=2}^{N} f_n(x)\right| = 1 \text{ for any } N\in\mathbb{N}.$$

Nevertheless, $\int_0^\infty f = \int_0^\infty \sum_{n=2}^{\infty} f_n = \sum_{n\geq 2}^{\infty}\int_0^\infty f_n = \sum_{n=2}^{\infty}\frac{1}{n^2}$, the integral converges.
But, for each $n\in\mathbb{N}$, $f(n+\frac{1}{n^2})=1$, so $f(x)$ does not tends to zero as $x\to\infty$.⌟

Assume that $f(x)$ is continuous uniformly
If f does not converges to zero, then there is $\varepsilon>0$ such that
for each $R>0$, there is $x\geq R$ such that $f(x)\geq\varepsilon$.
Now, let $\delta>0$ such that $|x-y|<\delta \Rightarrow |f(x)-f(y)|<\frac{\varepsilon}{2}$.
And, let $\{x_n\}$ be a positive sequence such that for all $n\in\mathbb{N}$,

$$x_{n+1}>x_n+2\delta \text{ and } f(x_n)\geq\varepsilon.$$

Then, $\int_0^{x_n+\delta} f \geq \sum_{k=2}^{n}\frac{\varepsilon}{2}\cdot 2\delta = n\varepsilon\delta \to\infty$ as $n\to\infty$,
So it contradicts to $\int_0^\infty f<\infty$. ☒

⌈2023 Spring.

13. Let α be a geodesic on the unit sphere $S^2=\{(x,y,z) \mid x^2+y^2+z^2=1\}$. Prove that α is contained in a plane.

Proof. Let $\alpha: I \to S^2$ be the geodesic. Then, the normal vector of S^2 at $\alpha(t)$ is $\alpha(t)$. And, since α is geodesic, by definition,

$$[n, T, T'] = [\alpha(t), \alpha'(t), \alpha''(t)/\kappa(t)] = \langle \underset{\sim}{\alpha} \times \underbrace{\alpha'}_{T}, \underbrace{\alpha''}_{\kappa N} \rangle / \kappa(t) = 0$$

Meanwhile, the torsion of α is

$$\tau(t) = \langle B', N \rangle = \langle \alpha' \times \alpha'' \cdot \tfrac{1}{\kappa}, \alpha'' \cdot \tfrac{1}{\kappa} \rangle = [\alpha'$$

$T \times N$

$(\alpha' \times \alpha'') \cdot \frac{1}{\kappa}$

$$\begin{pmatrix} & \kappa & \\ -\kappa & & \tau \\ & -\tau & \end{pmatrix}$$

B',

$B' = -\tau N$

「2023. Fall.

1. Suppose that $\{a_n\}$ and $\{b_n\}$ are bounded sequences of real numbers.

a). Prove that

$$\limsup_{n\to\infty}(a_n+b_n) \le \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n.$$

b). Give an example for the equality does not hold.

Proof

a). Since for each $n\in\mathbb{N}$,

$$\sup_{k\ge n}(a_n+b_n) \le \sup_{k\ge n} a_n + \sup_{k\ge n} b_n,$$

taking limit to the both sides we obtain the result.

b). Let $a_n=(-1)^n$ and $b_n=(-1)^{n+1}$. Then,

$\limsup(a_n+b_n)=0$

but

$\limsup a_n = \limsup b_n = 1.$

「2023 Fall.

2. Find all solutions to the ODE

$$xy' + y = x^2 \cdot y^2 \quad \text{and} \quad y(1) = 1.$$

Proof. Since $y(1) = 1 \neq 0$, $y > 0$ for some neighbor of $x = 1$. In this neibor,

$$xy' + y = x^2 y^2 \Rightarrow \frac{1}{x} \cdot \frac{1}{y^2} \cdot y' + \frac{1}{x^2} \cdot \frac{1}{y} = 1$$

$$\Rightarrow \left(\frac{1}{x} \cdot \frac{1}{y} \right)' = -1$$

$$\Rightarrow \frac{1}{xy} = -x + c$$

$$\Rightarrow \frac{-1}{x(x-c)} = y$$

Since $y(1) = 1$, $\frac{1}{c-1} = 1 \Rightarrow c = 2$.

Finally, $y(x) = \dfrac{1}{x(2-x)}$ for some neibor of $x = 1$.

is the solution of the initial value problem, we desired.

Moreover, by the uniqueness theorem, it is the unique solution through $(1,1) \in \mathbb{R}^2$,

$\sqrt{2}$2023. Fall.

4. Find all $\alpha > 0$ such that
$$\int_0^\infty \cos(x^\alpha)\, dx$$
exists as a real number.

Proof.

[2023 Fall.

6. Let $A \in M_{n\times n}(\mathbb{R})$ be given. Put $m_A(t) \in \mathbb{R}[t]$ be the minimal polynomials and $\chi_A(t) \in \mathbb{R}[t]$ be the Characteristic Polynomial.

a). Does there exists a 3 by 3 matrix A such that

$$m_A(t) = t^2 - t \quad \text{and} \quad \chi_A(t) = t^3 - t^2 \ ?$$

b). If $C \in M_{5\times 5}(\mathbb{R})$ such that $C^5 = I_5$, is it necessarily true that

$$m_C(t) = t^5 - 1 \ ?$$

Proof.

a). Let $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Then, $\chi_A(t) = t^2(t-1) = t^3 - t^2$.

And, $A\cdot(A-I) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$,

so $m_A(t) = t(t-1)$.

There fore, exists.

b). if $C = I_5$, then $C^5 = I_5$ but the minimal polynomial is $t-1$.
So, not necessary. ▯

[2023 Fall.

7. Let $K = \mathbb{F}_{p^2}$ be a finite field of order of p^2.

a). Find the number of all 3×3 invertible matrices over K.

b). Let V be a 5-dimensional vector space over K and $W \subset V$ be a 2-dimensional subspace. Find the number of all K-linear maps

$$T: V \to V \quad \text{such that } T[W] \subseteq W.$$

Proof a). Since $A \in M_{3\times3}(K)$ is invertible if and only if the rows of A are linearly independent, so the possible cases are

$$\underbrace{[(p^2)^3 - 1]}_{\substack{\text{Choose first row}\\ \text{except zero}}} \cdot \underbrace{[(p^2)^3 - p^2]}_{\substack{\text{Choose second}\\ \text{except multiple of}\\ \text{the first}}} \cdot \underbrace{[(p^2)^3 - p^2 \cdot p^2]}_{\text{similar.}}$$

So, $|GL_3(K)| = (p^6 - 1)(p^6 - p^2)(p^6 - p^4)$.

b). Let $\{\alpha_1, \alpha_2\}$ be the basis for W.

Since $T[W] \subseteq W \iff T(\alpha_1), T(\alpha_2) \in W$, so $T[W] \subseteq W$ iff

$T(\alpha_1) = a \cdot \alpha_1 + b \cdot \alpha_2$ and $T(\alpha_2) = c \cdot \alpha_1 + d \cdot \alpha_2$ for some $a, b, c, d \in K$

And, let $\{\alpha_1, \alpha_2, \cdots, \alpha_5\}$ be the extended basis for V.

$T(\alpha_i) = \sum_{j=1}^{5} a_j \alpha_j$ for $i \geq 3$, so the number of such a T is

$$p^2 \cdot p^2 \cdot p^2 \cdot p^2 \cdot ((p^2)^5)^3.$$

「2023, Fall.

14. If X is Lindelöf and Y is compact, then $X\times Y$ is Lidelöf. Prove.

proof. Let $\mathcal{O}$ be the open cover for $X\times Y$. Let $x_0\in X$ be fixed.
Since $\{x_0\}\times Y\to Y : (x_0,y)\mapsto y$ is Homeomorphism, $\{x_0\}\times Y\cong Y$ is compact.
Since $\mathcal{O}$ covers $X\times Y$, it also covers $\{x_0\}\times Y$, so there is a finite subcover $F_0\subset\mathcal{O}$, which satisfies $\{x_0\}\times Y\subset\bigcup F_0\subseteq X\times Y$
Given $y\in Y$, $(x_0,y)\in\bigcup F$, open, so there exist open sets $U_y\times V_y\subseteq X\times Y$ such that $(x_0,y)\in U_y\times V_y\subseteq\bigcup F$. The compactness gives that the open cover $\{U_y\times V_y \mid y\in Y\}$ of $\{x_0\}\times Y$ has a finite subcover

$\{U_{y_1}\times V_{y_1},\cdots,U_{y_n}\times V_{y_n}\}$. Let $W_0=U_{y_1}\cap\cdots\cap U_{y_n}$

Then, $x_0\in W_0$ since $x_0\in U_{y_i}$ for all $1\le i\le n$, so that $\{x_0\}\times Y\subset W_0\times Y$, and W_0 is open, being finite intersection of open sets. Moreover,

$W_0\times Y\subseteq\bigcup F$, because $(x,y)\in W_0\times Y \Leftrightarrow x\in U_{y_1}\cap\cdots\cap U_{y_n}$ and $y\in V_{y_k}\ \exists k$,
$\Rightarrow (x,y)\in U_{y_k}\times V_{y_k}\ \exists k$
$\Rightarrow (x,y)\in\bigcup F$.

Since $x_0\in X$ was arbitrary, we can consider open cover $\{W_x \mid x\in X\}$ for X, and since X is Lindelöf, it has a countable subcover $\{W_{x_k} \mid k\in\mathbb{N}\}$.
So, $X\times Y=\left(\bigcup_{k\in\mathbb{N}}W_{x_k}\right)\times Y=\bigcup_{k\in\mathbb{N}}W_{x_k}\times Y\subseteq\bigcup_{k\in\mathbb{N}}\bigcup F_k$,
where $F_k\subset\mathcal{O}$ is the finite subcover determined when $x_k\in X$ chosen, thus $\mathcal{O}$ has a countable subcover ∎

「Debris

Let $\mathcal{O}=\{U_\lambda \mid \lambda\in\Lambda\}$ be an open cover of $X\times Y$.
Given $x_0\in X$, since the projection $\{x_0\}\times Y\to Y : (x_0,y)\mapsto y$ is Homeomorphism, so $\{x_0\}\times Y\cong Y$ is compact. Therefore, there is a finite subcollection

$\mathcal{O}_{x_0}\subseteq\mathcal{O}$ satisfying $\{x_0\}\times Y\subseteq\bigcup\mathcal{O}_{x_0}$.

For each $x\in X$, define $W_x=\bigcup\mathcal{O}_x$. Let $x_0\in X$ be fixed.
Since W_{x_0} is open, being union of open sets, so given $(x,y)\in W_{x_0}$, there exist open sets $U_x\subseteq X$ and $V_x\subseteq Y$ such that

$$(x,y)\in U_x\times V_x\subseteq W_{x_0}$$

Then, $\{U_x\times V_x \mid x\in W_{x_0}\}$ is an open cover of $\{x_0\}\times Y$, so there is finite subcover $\{U_{x_0}^1\times V_{x_0}^1,\cdots,U_{x_0}^n\times V_{x_0}^n\}$. Take $A_{x_0}=U_{x_0}^1\cap\cdots\cap U_{x_0}^n$.
Then, since $x_0\in A_{x_0}$, it is non-empty open set satisfying

$$\{x_0\}\times Y\subseteq A_{x_0}\times Y\subseteq W_x$$

「2024, Spring.

1. Let $A, B \in M_{m\times n}(F)$. Show that $\operatorname{rank}(A+B) \le \operatorname{rank} A + \operatorname{rank} B$. $\boxed{5m}$

Proof. By definition,

$$\operatorname{rank}(A+B) = \dim \operatorname{Im}(A+B) \text{ and } \operatorname{rank} A = \dim \operatorname{Im} A,\ \operatorname{rank} B = \dim \operatorname{Im} B.$$

Given $\alpha \in \operatorname{Im}(A+B)$, $\alpha = (A+B)\beta = A\beta + B\beta \in \operatorname{Im} A + \operatorname{Im} B$ for some $\beta \in F^m$, therefore $\operatorname{Im}(A+B) \subseteq \operatorname{Im} A + \operatorname{Im} B$. Now,

$$\operatorname{rank}(A+B) \le \dim(\operatorname{Im} A + \operatorname{Im} B) \le \dim \operatorname{Im} A + \dim \operatorname{Im} B = \operatorname{rank} A + \operatorname{rank} B.$$

2. Suppose that P and Q are positive definite such that $P^2 = Q^2$. Show $P = Q$.

Proof. Since P is Hermitian, there exists an orthonormal basis $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$ such that $P(\alpha_i) = c_i\alpha_i$ for some $c_i > 0$, for each i.

Let $1 \le i \le n$ be fixed. Then, by the assumption, $\boxed{15m}$

$$P^2\alpha_i = c_i^2\alpha_i = Q^2\alpha_i \Rightarrow (Q^2 - c_i^2 I)\alpha_i = (Q - c_i I)(Q + c_i I)\alpha_i = 0.$$

Since Q is positive definite, $(Q - c_i I)\alpha_i = 0$, because; if it is not zero, then

$$(Q - c_i I)(Q + c_i I)\alpha_i = (Q + c_i I)(Q - c_i I)\alpha_i = 0,$$

So $-c_i$ is negative eigenvalue of Q, it contradicts to Q is positive definite. Therefore, for each $1 \le i \le n$,

$$P\alpha_i = c_i\alpha_i = Q\alpha_i,$$

and since $\mathcal{B}$ is basis, therefore $P = Q$. ∎

「2024, Spring

3. Let $T : \mathbb{R}^5 \to \mathbb{R}^5 ; x \mapsto Ax$ where

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

a). There exists T-invariant subspace $V \subseteq \mathbb{R}^5$ s.t $\dim V = 4$?

b). 〃 s.t $\dim V = 2$, $T|_V$ is diagonalizable?

c) 〃 $\dim V = 3$, 〃

Proof.

a). Let $V = \langle e_1, e_2, e_3, e_4 \rangle$. Then,

$$Ae_1 = e_1,\ Ae_2 = e_1 + e_2,\ Ae_3 = e_3,\ Ae_4 = e_3 + e_4$$

are all contained in V, so $T[V] \subseteq V$. (clearly $\dim V = 4$)

b). Let $V = \langle e_1, e_3 \rangle$. Then $Ae_1 = e_1$, $Ae_3 = e_3$ are contained in V, and

$$[T|_V]_{\{e_1, e_3\}} = I_2.$$ So $T|_V$ is diagonalizable.

c). Observe characteristic polynomial of T is $(t-1)^5$.
If such a subspace V exists, then the characterstic polynomial of $T|_V$ is $(t-1)^3$. Since $T|_V$ is diagonalizable if and only the minimal polynomial splits and separable, thus the minimal polynomial of $T|_V$ must be $(t-1)$.
That is, $T|_V = I_V$ but it implies that $\ker(T-I)$ has dimension more than or equal 3, it contradicts to

$$\dim \ker (A - I) = \dim \ker \begin{pmatrix} 0 & 1 & & & \\ & 0 & 0 & & \\ & & 0 & 1 & \\ & 0 & & 0 & 1 \\ & & & & 0 \end{pmatrix} = 2.$$

「2024, Spring.

5. Let K be a field, and $L_1, L_2 \subseteq \overline{K}$ be finite extensions of K. 41

a). Prove $L = L_1L_2$ is finite extension.

b). If L_1 and L_2 are Galois over K, then L/K is Galois.

a). Since L_1 and L_2 both are finite, denote

$$L_1 = K(\alpha_1, \cdots, \alpha_n) \quad \text{and} \quad L_2 = K(\beta_1, \cdots, \beta_r)$$ where $\alpha_i, \beta_j \in \overline{K}$ are algebraic over K.

Now, $$L = L_1L_2 = \bigcap_{\substack{E \leq \overline{K} \\ L_1, L_2 \leq E}} E = K(\alpha_1, \cdots, \alpha_n, \beta_1, \cdots, \beta_r)$$

is finitely generated by algebraic elements α_i, β_j, so L is finite. ㅈㅁ.

b). Since L_1 and L_2 are splitting field of K, let f_1 and f_2 be polynomials over K and such that L_i is the splitting field of f_i, $i = 1, 2$.
It is possible, because L_1, L_2 both are finite extension.
Now let E be the splitting field of f_1f_2. Then,
$E \subseteq L_1L_2$, because E is the smallest field containing all roots of f_1f_2,
$L_1L_2 \subseteq E$ because E contains all roots of f_1 and f_2. So, $L = E_1E_2$,
therefore L is splitting field over K. ㅈㅁ.

Since L_1 and L_2 both are Galois, for some separable f_1 and f_2 over K, L_1 and L_2 are splitting field of f_1 and f_2 respectively.

L_1L_2 가 f_1f_2 의 splitting field => ok
근데 f_1f_2가 separable 이란 보장 X.
중복 linear factor 소거 => f_1f_2가 여전히 over K 인가?

Revised in the Algebra/Field Theory/

「2024 Spring.

7. Let $f:(-1,1)\to\mathbb{R}$ be a differentiable function satisfying $f(0)=0$ and let $g:(-1,1)\to\mathbb{R}:\begin{cases} f(x)/x & \text{if } x\neq 0 \\ f'(0) & \text{if } x=0.\end{cases}$

a). If $f\in C^2$, then $g\in C^1$ (Prove)

b). If $f\in C^1$ but $\notin C^2$, then $g\in C^1$ may not hold.

Proof. Notice that g is continuous, since

$$\lim_{x\to 0} g(x) = \lim_{x\to 0}\frac{f(x)-f(0)}{x-0} = f'(0).$$

a) For $x\neq 0$,

$$g'(x) = \left(\frac{f(x)}{x}\right)' = \frac{xf'(x)-f(x)}{x^2}$$ exists. By the L'Hospital Theorem,

$$\lim_{x\to 0} g'(x) = \lim_{x\to 0}\frac{f'(x)+xf''(x)-f'(x)}{2x} = \frac{1}{2}\cdot\lim_{x\to 0} f''(x) = \frac{1}{2}f''(0).$$

(Since $x^2\to 0$ and $xf'(x)-f(x)\to 0$ as $x\to 0$, we can use the theorem).

And, $$g'(0) = \lim_{x\to 0}\frac{g(x)-g(0)}{x-0} = \lim_{x\to 0}\frac{f(x)/x-f'(0)}{x-0} = \lim_{x\to 0}\frac{f(x)-xf'(0)}{x^2} = \frac{1}{2}f''(0).$$

Since the Lopital theorem again, so $g\in C^1$.

「2024 spring.

8. Let $C_b(\mathbb{R})$ be the set of all real-valued bounded continuous functions on $\mathbb{R}$.
For two functions $f, g \in C_b(\mathbb{R})$, define
$$d(f,g) := \sup_{x\in\mathbb{R}} |f(x) - g(x)|.$$
Let $\{f_n\}_{n=1}^{\infty}$ be a sequence in $C_b(\mathbb{R})$ such that:
$\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $m,n \geq N \Rightarrow d(f_m, f_n) < \varepsilon$.

a). Prove that there exists $f \in C_b(\mathbb{R})$ such that $\lim_{n\to\infty} d(f, f_n) = 0$

b). Prove that $\lim_{n\to\infty} \left(\sup_{x\in\mathbb{R}} |f_n(x)| \right) = \sup_{x\in\mathbb{R}} |f(x)|$.

Proof. For each $x \in \mathbb{R}$, a sequence $\{f_n(x)\}$ in $\mathbb{R}$ is Cauchy sequence by hypothesis, Since $\mathbb{R}$ is complete, $\lim_{n\to\infty} f_n(x)$ exists, Put this $f(x) \in \mathbb{R}$.

a).

Since $x \in \mathbb{R}$ was arbitrary, $f(x)$ is well-defined on $\mathbb{R}$, but still not necessarly $f \in C_b(\mathbb{R})$, and $d(f, f_n) \to 0$.

Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $m > n \geq N$ implies that $\sup_{x\in\mathbb{R}} |f_m(x) - f_n(x)| \leq \varepsilon$.
So, for any $x \in \mathbb{R}$, $|f_m(x) - f_n(x)| \leq \varepsilon$. Letting $m \to \infty$, we obtain $|f(x) - f_n(x)| \leq \varepsilon$.
Since x was arbitrary, $\sup_{x\in\mathbb{R}} |f(x) - f_n(x)| \leq \varepsilon$, for $n \geq N$.
Therefore, the triangle inequality gives $|f(x)| \leq |f_n(x)| + \varepsilon$ for all $x \in \mathbb{R}$ and since f_n was bounded, f is bounded. And, the argument above proves $\sup_{x\in\mathbb{R}} |f - f_n| \to 0$ as $n \to \infty$, So since f_n is continuous and

$f_n \to f$ uniformly, so f is also continuous, so $f \in C_b(\mathbb{R})$ and a) is proved.

b). Since $\left| \sup_{x\in\mathbb{R}} |f_n(x)| - \sup_{x\in\mathbb{R}} |f(x)| \right| \leq \sup_{x\in\mathbb{R}} |f_n(x) - f(x)| \to 0$ as $n \to \infty$,

Proved directly.

[2024 Spring

12. Compute the Gauss curvature at $p=(0,0,0)$ of the smooth surface given by $z=y^2-x^2$.

Proof. Consider the Monge patch of the given surface

$$X: \mathbb{D} \to \mathbb{R}^3 : (u,v) \mapsto (u, v, v^2-u^2)$$

Then, $X_u=(1,0,-2u)$ and $X_v=(0,1,2v)$, so the first fundamental form

$E=1+4u^2$, $F=-2uv$, $G=1+4v^2$

so $\det\begin{pmatrix} E & F \\ F & G \end{pmatrix} = 1+4u^2+4v^2+16u^2v^2-4u^2v^2 = 1$ at $(u,v)=(0,0)$

and $\begin{pmatrix} E & F \\ F & G \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ at $(u,v)=(0,0)$.

Meanwhile, the normal of the given surface is

$X_u \times X_v = (2u, -2v, 1)$ gives $n=(0,0,1)$ at $(u,v)=(0,0)$.

Meanwhile, $X_{uu}=(0,0,-2)$, $X_{uv}=(0,0,0)$, $X_{vv}=(0,0,2)$ so

$L_{uu}=-2$, $L_{uv}=0$, $L_{vv}=2$.

$L_u^u = L_{uu} g^{uu} + L_{vu} \cdot g^{vu} = 2$

$L_u^v = L_v^u = L_{uu} \cdot g^{uv} + L_{vu} \cdot g^{vv} = -4$

$L_v^v = L_{uv} g^{uv} + L_{vv} \cdot g^{vv} = 4$.

Therefore, the Gaussian curvature is

$$\det\begin{pmatrix} 2 & -4 \\ -4 & 4 \end{pmatrix} = 24.$$

Note: Gaussian curvature is $\det II / \det I$.

[2024. spring.

14. Let X be a compact hausdorff space. Prove that

X is metrizable if and only if X has a countable basis.

Proof Suppose that X is metrizable by the metric $d: X\times X \to \mathbb{R}$.
Given $n \in \mathbb{N}$, $\{B_{\frac{1}{n}}(x) \mid x \in X\}$ is an open cover of X, so the compactness gives that $\mathcal{B}_n = \{B_{\frac{1}{n}}(x_i) \mid i \in F_n\}$, the finite subcover exists with finite index set F_n. Now, put $\mathcal{B} = \bigcup_{n\in\mathbb{N}} \mathcal{B}_n$.

Let $x \in X$ be given and let $B_r(y)$ be an open ball containing x.
Then, for $\varepsilon = r - d(x,y)$, $x \in B_\varepsilon(x) \subseteq B_r(y)$, because

$a \in B_\varepsilon(x) \Leftrightarrow d(a,x) < \varepsilon = r - d(x,y) \Rightarrow d(a,y) = d(a,x) + d(x,y) < r$.

Now, let $N \in \mathbb{N}$ such that $\frac{1}{N} < \frac{\varepsilon}{2}$. Then, since each $\mathcal{B}_n$ covers X, for some x_i, $x \in B_{\frac{1}{N}}(x_i)$, and

$z \in B_{\frac{1}{N}}(x_i) \Leftrightarrow d(z,x_i) < \frac{1}{N} \Rightarrow d(z,x) \leq d(z,x_i) + d(x_i,x) < \frac{1}{N} + \frac{1}{N} < \varepsilon$.

So, $x \in B_{\frac{1}{N}}(x_i) \subseteq B_\varepsilon(x)$, it proves $\mathcal{B}$ is a basis for (X,d),

and since $\mathcal{B}$ is countable union of finite sets, so it is countable.

Conversely, since compact Hausdorff space is normal, so it is regular.
By the Urysohn metrization theorem, second countable regular space is metrizable, so proved.

2024, Fall.

1. Let V be a finite-dimensional vector space over a field F. $\boxed{7m}$
Let $T: V \to V$ be a linear operator such that $T^2 = T$. Prove
a). $V = \ker T \oplus \mathrm{Im}\, T$
b). T is diagonalizable.

Proof. a). Given $\alpha \in V$, $\alpha = \alpha - T(\alpha) + T(\alpha)$, $T(\alpha) \in \mathrm{Im}\, T$ is trivial,
and $T(\alpha - T(\alpha)) = T(\alpha) - T^2(\alpha) = (T - T^2)(\alpha) = 0$, so $\alpha - T(\alpha) \in \ker T$.
Since α was arbitrary, $V = \ker T + \mathrm{Im}\, T$.
Moreover, let $\alpha \in \ker T \cap \mathrm{Im}\, T$. Then, $T(\alpha) = 0$ and $\alpha = T(\beta)$ for some $\beta \in V$. Now, $\alpha = T(\beta) = T^2(\beta) = T(T(\beta)) = T(\alpha) = 0$, so $\ker T \cap \mathrm{Im}\, T = \{0\}$,
it proves $V = \ker T \oplus \mathrm{Im}\, T$.

b). Since $T^2 - T = 0$, the minimal polynomial m of T divides $x^2 - x$.
If $m = x$ or $x - 1$, then $T \equiv 0$ or $T = \mathrm{Id}$, already diagonal.
If $m = x(x-1)$, then it splits over the given field and separable,
so T is diagonalizable. $\square$

Note: Naturally, $T|_{\mathrm{Im}\, T} = \mathrm{Id}$, so a) implies b).

「2024 Fall.

8. Solve the followings ; $x \in \mathbb{R}$,

58

a). $x''(t) - 2 \cdot x'(t) + x(t) = 0$

b). $x''(t) - 2x'(t) + x(t) = te^t + 4$.

Proof.

a). Since the characteristic polynomial of the equation is

$$r^2 - 2r + 1 = (r-1)^2,$$

the general solution is

$$x_g(t) = C_1 e^t + C_2 \cdot te^t \quad \text{where } C_1, C_2 \in \mathbb{R}.$$

b). Calculate the Wronskian ;

$$W(e^t, te^t) = \begin{vmatrix} e^t & te^t \\ e^t & e^t + te^t \end{vmatrix} = e^{2t} + te^{2t} - te^{2t} = e^{2t}.$$

Therefore, the Particular solution is

$$Y(t) = \int_{t_0}^{t} \begin{vmatrix} e^u & e^t \\ ue^u & te^t \end{vmatrix} \cdot b(u) / W(e^u, ue^u)\, du$$

$$= \int_{t_0}^{t} (e^u te^t - ue^u e^t) \cdot (ue^u + 4) / e^{2u}\, du$$

$$= \int_{t_0}^{t} (ue^{2u} te^t + 4 e^u te^t - u^2 e^{2u} e^t - 4u e^u e^t) / e^{2u}\, du$$

$$= \int_{t_0}^{t} u \cdot te^t + 4 \cdot te^t \cdot e^{-u} - u^2 \cdot e^t - 4e^t \cdot ue^{-u}\, du$$

Now, $x_g + Y$ is the general solution.

「2024, Fall,

9. For a fixed $\xi \in \mathbb{R}$, let
$$f(z) = \frac{e^{-2\pi i z \xi}}{\cosh(\pi z)} \qquad (z \in \mathbb{C})$$

a). Show that $\alpha = i/2$ and $\beta = 3i/2$ are simple poles of f

b). Compute the residues of f at α and β.

c). Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{\cosh(\pi x)}\,dx$.

Proof. a). Notice that $e^{-2\pi i x \xi}$ is entire and has no zero, we need to investigate pole of $1/\cosh(\pi z)$.

Since $\cosh(\pi z_0) = 0$ if $z_0 = \alpha$ or β, and
$$\lim_{z \to z_0} \frac{\cosh(\pi z) - \cos(\pi z_0)}{z - z_0} = \sinh(\pi z_0)\cdot \pi \neq 0,$$
so
$$\lim_{z \to z_0} (z - z_0)^1 \cdot \frac{1}{\cosh(\pi z)} = \frac{1}{\pi \sinh(\pi z_0)}$$
exists, so α and β are simple poles.

b). Since α and β both are simple pole,
$$\operatorname{Res}(f, \alpha) = \lim_{z \to \alpha} (z - \alpha)\cdot f(z) = e^{-2\pi i \alpha \xi} \cdot \frac{1}{\pi \cdot \sinh(\pi \alpha)} = \frac{e^{\pi \xi}}{\pi \cdot \sinh(\pi i/2)} = \frac{e^{\pi \xi}}{\pi i}.$$
$$\operatorname{Res}(f, \beta) = e^{-2\pi i \beta \xi} \cdot \frac{1}{\pi \cdot \sinh(\pi \beta)} = \frac{e^{3\pi \xi}}{\pi \cdot (-i)} = -\frac{e^{3\pi \xi}}{\pi i}.$$

c). Let C_R be the rectangle $C_R = C_1 * C_2 * C_3 * C_4$ along four line segment $-R$ to R, R to $R+i$, $R+i$ to $-R+i$, $-R+i$ to $-R$.

And, observe that
$$f(x+i) = \frac{e^{-2\pi i x \xi + 2\pi \xi}}{\cosh(\pi x + \pi i)} = e^{2\pi \xi} \cdot - f(x)$$
Therefore, $\displaystyle\int_{C_1 + C_3} f = \left(\int_{-R}^{R} f(x)\,dx\right)\cdot\left(1 + e^{2\pi \xi}\right).$

And, Since $\left|f(R+it)\right| = \left|\dfrac{e^{-2\pi i R \xi + 2\pi t \xi}}{\cosh(\pi R + \pi i t)}\right| \to 0$ as $\left|\cosh(\pi R + \pi i t)\right| \to \infty$ as $R \to \infty$.

Therefore, $\int_{C_2} f$, $\int_{C_4} f \to 0$, so, by the Residue theorem,
$$\int_{-\infty}^{\infty} f(x)\,dx = \frac{1}{1 + e^{2\pi \xi}} \cdot \lim_{R \to \infty} \int_{C_R} f = \frac{1}{1 + e^{2\pi \xi}} \cdot 2e^{\pi \xi}.$$

「2024 Fall.

12. Let $p: X \to Y$ be a quotient map.

Show that if Y is connected and $p^{-1}[\{y\}]$ is connected for each $y \in Y$, then X is connected.

Proof. Suppose that X is disconnected, with a separation $\{\mathcal{U}, V\}$.

Given $y \in Y$, $p^{-1}[\{y\}] \subseteq X$ is connected, so either

$$p^{-1}[\{y\}] \subseteq \mathcal{U} \text{ or } p^{-1}[\{y\}] \subseteq V.$$

(If not, then $\{\mathcal{U}, V\}$ is a separation of $p^{-1}[\{y\}]$)

Meanwhile, observe that

$$Y = P[X] = P[\mathcal{U} \cup V] = P[\mathcal{U}] \cup P[V]$$

Let $a \in P[\mathcal{U}]$ be given. Then, $a = P(b)$ for some $b \in \mathcal{U}$.

And, $P^{-1}[\{a\}]$ contains b, so $p^{-1}[\{a\}] \cap \mathcal{U} \neq \emptyset$, therefore $P^{-1}[\{a\}] \subseteq \mathcal{U}$

Thus, $p^{-1}[P[\mathcal{U}]] \subseteq \mathcal{U}$ and $\mathcal{U} \subseteq P^{-1}[P[\mathcal{U}]]$ generally, so

$\mathcal{U} = P^{-1}[P[\mathcal{U}]]$ is open. Since P is quotient map, $P^{-1}[P[\mathcal{U}]] = \mathcal{U}$ is open implies that $P[\mathcal{U}]$ is also open set, similarly $P[V]$ is open

Moreover, $P[\mathcal{U}] \cap P[V] = \emptyset$, because suppose that

for some $a \in Y$, $a = P(b) = P(c)$ for some $b \in \mathcal{U}$ and $c \in V$.

Then, $p^{-1}[\{a\}] = p^{-1}[P[\{b\}]] = P^{-1}[P[\{c\}]]$, but

Since $b \in p^{-1}[P[\{b\}]] \cap \mathcal{U}$ and $c \in P^{-1}[P[\{c\}]] \cap V$,

$p^{-1}[\{a\}] \subseteq \mathcal{U}$ and V, but it contradicts to $\mathcal{U} \cap V = \emptyset$.

So, $\{P[\mathcal{U}], P[V]\}$ is a separation of Y, it is contradiction ⊜

⌈2024 Fall.

13. Let X be any space and Y be Compact Hausdorff space.

$f: X \to Y$ is Continuous map if and only if the graph $G_f = \{(x, f(x)) \mid x \in X\}$ is closed.

T_2

Compact

proof. Let $f: X \to Y$ be a Continuous map.

Let $(x,y) \in X \times Y \setminus G_f$ be given. Then, $y \neq f(x)$, so there exist disjoint open sets $\mathcal{U}, V$

$$\mathcal{U} \cap V = \phi, \quad y \in \mathcal{U}, \quad f(x) \in V$$

Since $f(x) \in V \overset{\text{def}}{\iff} x \in f^{-1}[V]$ and f is Continuous, $f^{-1}[V]$ is an open set in X.

now, $(x,y) \in f^{-1}[V] \times \mathcal{U}$

And, $(f^{-1}[V] \times \mathcal{U}) \cap G_f = \phi$, because if $(x, f(x)) \in f^{-1}[V] \times \mathcal{U}$, then

$f(x) \in V$ and $f(x) \in \mathcal{U}$, it is contradiction to disjointness.

Therefore, $(x,y) \in f^{-1}[V] \times \mathcal{U} \subseteq X \times Y \setminus G_f$, So proved. $\lrcorner \Rightarrow$

Conversely, let G_f is closed in $X \times Y$, under the Product Topology.

Let $C \subseteq Y$ be a Closed set in Y. Then, $G_f \cap (X \times C)$ is closed. Moreover,

$$P_X[G_f \cap (X \times C)] = \{x \in X \mid f(x) \in C\} = f^{-1}[C]$$

is closed, because $P_X : X \times Y \to X : (x,y) \mapsto x$ is closed map if Y is Compact. ☒

Note: for $g = f \times i : X \times Y \to Y \times Y : (x,y) \mapsto (f(x), y)$,

$$g^{-1}[\Delta(Y)] = \{(x,y) \in X \times Y \mid (f(x), y) \in \Delta(Y)\} = G_f \lrcorner \Leftarrow$$

「2024. Fall.

14. Let $S \subset \mathbb{R}^3$ be a smooth surface and $p \in S$ be a point.

$\ulcorner$2025. Spring.

1. Let F be any field, and let $V = M_{n\times n}(F)$, and let
$$T: V \to V : A \mapsto A + A^t$$

a). Show that $V = \ker T \oplus \operatorname{Im} T$.

b). Is T diagonalizable? Find all eigenvalues of T with multiplicities.

Proof. Let $ch(F) \neq 2$.

a). Given $A \in V$,
$$A = \frac{1}{2}(A - A^t) + \frac{1}{2}(A + A^t)$$
And, since $-\left(\frac{1}{2}(A-A^t)\right)^t = -\frac{1}{2}(A^t - A) = \frac{1}{2}(A - A^t)$ implies $\frac{1}{2}(A-A^t) \in \ker T$ and $\frac{1}{2}(A+A^t) = T(\frac{1}{2}A) \in \operatorname{Im} T$, so $V = \ker T + \operatorname{Im} T$.
Meanwhile, given $A \in \ker T \cap \operatorname{Im} T$,
$$A + A^t = 0 \quad \text{and} \quad A = B + B^t \text{ for some } B \in V.$$
But, $0 = A + A^t = B + B^t + B^t + B = 2\cdot(B + B^t) = 2\cdot A$,
so $A = 0$. It proves $\ker T \cap \operatorname{Im} T = \{0\}$, thus $V = \ker T \oplus \operatorname{Im} T$.

b). Since $A \in \ker T \Leftrightarrow A = -A^t$, so, all diagonal entries are zero and if the upper entries are chosen, then the lower entries are determined. So $(n^2 - n)/2$ is the dimension of $\ker A$.

Meanwhile, given $A \in \operatorname{Im} T$, $A = B + B^t$ for some $B \in V$ and
$$T(A) = T(B) + T(B^t) = B + B^t + B + B^t = 2\cdot A.$$
So, $\operatorname{Im} T \subseteq \ker(T - 2I)$. Conversely, given $A \in \ker(T - 2I)$, $A + A^t - 2A = A^t - A = 0$, so $A = A^t$. Meanwhile,
$$T(\tfrac{1}{2}A) = \tfrac{1}{2}A + \tfrac{1}{2}A^t = A, \text{ so } A \in \operatorname{Im} T.$$ Thus, $\operatorname{Im} T = \ker(T - 2I)$.
By the dimension theorem,
$$\dim \operatorname{Im} T = \frac{n^2 + n}{2}.$$
Therefore, 0 and 2 are eigenvalues of T with multiplicity $\frac{n^2-n}{2}, \frac{n^2+n}{2}$ respectively, and diagonalizable. $\boxtimes$

「2025 Spring.

3.

a). Prove that a polynomial ring $F[x]$ over a field F is P.I.D.

b) Find $f(x) \in \mathbb{C}[x]$ such that $(f(x)) = (x^5+2x^4+3x^3+2x^2+x, x^6-1)$.

Proof.

a). Let $I \subseteq F[x]$ be an ideal. If $I = F[x]$ or $I = \{0\}$, then $I = (1)$ or (0).
Now, assume that $\{0\} \subsetneq I \subsetneq F[x]$. Consider a map

$$\deg : F[x] \to \mathbb{N} \cup \{0\} : f(x) \mapsto \deg f(x)$$

Let $m = \min\{\deg[I \setminus \{0\}]\}$, it is guaranteed by the Well-ordering Principle, and $m \geq 1$, because I cannot contain a unit, by the assumption.
Now, choose $P(x) \in \deg^{-1}[\{m\}]$ Now, by the division algorithm,

$$f(x) = P(x)q(x) + r(x) \text{ for some } q, r \in \mathbb{C}[x], \text{ with } r=0 \text{ or } \deg r < \deg p$$

But, since I is Ideal, $r(x) = f(x) - P(x)q(x) \in I$, so the minimality of degree of P gives that $r=0$, so $f = Pq \in I$. Therefore, $I \subseteq (P(x))$.
And, $(P(x)) \subseteq I$ is trivial, so $(P(x)) = I$.

b). Since $x^6 - 1 = (x-1)(x^5+x^4+x^3+x+1) = (x-1)(x^2+x+1)(x^3+1)$
and $x^5+2x^4+3x^3+2x^2+x = x(x^4+2x^3+3x^2+2x+1) = x(x^2+x+1)^2$
so, the greatest common divisor is (x^2+x+1), so x^2+x+1 is the generating polynomial we desired, because $(g,h) = (\gcd(g,h))$ in P.I.D. ∎

「Debris

b). By the division,

$$x^6 - 1 = (x^5+2x^4+3x^3+2x^2+x)(x-2) + x^4+4x^3+3x^2+2x$$

$$x^5+2x^4+3x^3+2x^2+x = (x^4+4x^3+3x^2+x)(x-2) + 8x^3+6x^2+5x$$

※ 생각을 해보자, $\mathbb{C}$ 는 Algebraic Closure 이고, 즉 각 Polynomial 들은 linear factor 로 쪼개진다. 즉, 나머지정리는 5? 6? 번은 써야하는데,
~~사실~~, 3번인데 그건 아닌거아냐
차라리 공통 근을 찾는게 더 이상적.

Let $I \subseteq F[x]$ be an ideal. If $I = F[x]$, then $(1) = I$.
If $I = \{0\}$, then $I = (0)$. Now, assume I is non-trivial proper Ideal.
Let $P(x) \in I \setminus \{0\}$ be given with $\deg P(x) \leq \deg h(x)$ for all $h(x) \in F[x] \setminus \{0\}$.
(It is possible by well-ordering Principle)
Now, let $f(x) \in I$ be given. Then, by the division algorithm,

[2025 Spring.

6. Prove or disprove that

If $f:[0,1]\to\mathbb{R}$ is uniformly continuous, then it is Lipschitz continuous.

Proof. Let $f:[0,1]\to\mathbb{R}: x\mapsto\sqrt{x}$. Since f is continuous and $[0,1]$ is compact, thus f is continuously uniformly. But, given $L>0$,

$|f(0)-f(x)|=|\sqrt{x}|>L\cdot|x|$ when $x<\frac{1}{L^2}$

Therefore, f does not satisfies the Lipschitz condition, so the given statement is false. ▨

Note: Given $P>0$, Consider $f(x)=x^P$. $(x\geq 0$

If $0<P\leq 1$, assume that $x\geq y\geq 0$. Put $h=x-y\geq 0$. Then, we want to show

$$x^P-y^P=(y+h)^P-y^P\leq h^P=(x-y)^P.$$

Given $\varepsilon>0$, take $\delta=$. Then, for all $x,y\in[0,1]$,

$|x-y|<\delta=\varepsilon \Rightarrow |f(x)-f(y)|\leq|\sqrt{x}-\sqrt{y}|=\frac{|x-y|}{\sqrt{x}+\sqrt{y}}$

[2025. Spring.

8. Let $\mathcal{U} = \{ z \in \mathbb{C} \mid |z| = 1 \}$ be the unit circle in the complex plane. Prove or disprove that:

Every continuous function on $\mathcal{U}$ can be approximated uniformly

by polynomials in the variable $z \in \mathbb{C}$.

Proof. Let $f : \mathcal{U} \to \mathbb{C} ; z \mapsto 1/z$ be a continuous map.
Directly, for a curve $C : t \mapsto e^{it}$ $(0 \le t \le 2\pi)$,

$$\oint_C \frac{1}{z} dz = \int_0^{2\pi} \frac{1}{e^{it}} ie^{it} = 2\pi i.$$

Now, suppose that there is a sequence of polynomials $\{P_n\}$ converges uniformly to f. Then, for each $n \in \mathbb{N}$, P_n is entire and so

$$\int_C P_n = 0$$ by the Cauchy-Goursat theorem,

therefore

$$\int_C f = \int_C \lim_{n\to\infty} P_n = \lim_{n\to\infty} \int_C P_n = 0$$

But it contradicts to $\int_C f = 2\pi i$. ☒

「2025 Spring.

13. Let D^2 be a closed unit disc in $\mathbb{R}^2$ centered at the origin.

a). Show that the quotient space $\mathbb{R}^2/D^2$ is homeomorphic to $\mathbb{R}^2$

b). Let $U = \text{int } D$. Determine $\mathbb{R}^2/U \cong \mathbb{R}^2$.

Proof. Let $\sim$ be a relation defined on $\mathbb{R}^2$ generated by:

$$(x_1, y_1) \sim (x_2, y_2) \iff (x_1, y_1), (x_2, y_2) \in D^2.$$

Then, it is clearly equivalent relation, and $\mathbb{R}^2/\sim = \mathbb{R}^2/D^2$, by definition.

a). Define a map

$$f: \mathbb{R}^2 \to \mathbb{R}^2 : (x, y) \mapsto \begin{cases} (0, 0) & \text{if } x^2+y^2 \le 1 \\ \left(1 - \frac{1}{\sqrt{x^2+y^2}}\right) \cdot (x, y) & \text{if } x^2+y^2 > 1 \end{cases}$$

Then, given $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$,

$$(x_1, y_1) \sim (x_2, y_2) \iff (x_1, y_1), (x_2, y_2) \in D^2 \text{ or } (x_1, y_1) = (x_2, y_2)$$

$$\iff f(x_1, y_1) = f(x_2, y_2)$$

and, it is onto continuous map, so f induces a Homeomorphism

$$\tilde{f}: \mathbb{R}^2/D^2 \to \mathbb{R}^2 : [(x, y)] \mapsto f(x, y).$$

「2025. Spring.

14. For $a, b, c > 0$, Consider the ellipsoid

$$E_{a,b,c} = \left\{ (x,y,z) \in \mathbb{R}^3 \,\middle|\, \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 = 1 \right\}.$$

a). If $a=b$, then $E_{a,a,c}$ is a surface of revolution (Prove!.)

b) Give a parametrization of $E_{a,a,c}$. Compute the first fundamental form of this parametrization and the Christoffel symbols

c). Show that the curve $E_{a,b,c} \cap \{x=0\}$ is the image of a geodesic.

a). Let

$$X : (0,2\pi) \times (0,2\pi) \to \mathbb{R}^3 ; (t,\theta) \mapsto (a \cdot \cos t \cdot \cos\theta,\ a \cdot \cos t \cdot \sin\theta,\ c \cdot \sin t)$$

It covers $E_{a,a,c}$ excepts a line, so $E_{a,a,c}$ is a surface of revolution.

b). The parametrization X is gave in a). And,

$X_t = (-a \cdot \sin t \cdot \cos\theta,\ -a \cdot \sin t \cdot \sin\theta,\ c \cdot \cos t)$

$X_\theta = (-a\cos t \cdot \sin\theta,\ a \cdot \cos t \cdot \cos\theta,\ 0)$. So the first fundamental form is

$E = a^2 \sin^2 t + c^2 \cos^2 t,\quad F = 0,\quad G = a^2 \cdot \cos^2 t.\quad \begin{bmatrix} E & F \\ F & G \end{bmatrix}^{-1} = \begin{bmatrix} E^{-1} & 0 \\ 0 & G^{-1} \end{bmatrix},$

$X_{tt} = (-a\cos t \cdot \cos\theta,\ -a\cos t \cdot \sin\theta,\ -c \cdot \sin t)$

$X_{t\theta} = (+a\sin t \sin\theta,\ -a\sin t\cos\theta,\ 0)$

$X_{\theta\theta} = (-a \cdot \cos t \cdot \cos\theta,\ -a\cos t \cdot \sin\theta,\ 0)$. Now,

$\Gamma^t_{tt} = \langle X_{tt}, X_t \rangle g^{tt} + \langle X_{tt}, X_\theta \rangle g^{\theta t} = (a^2 - c^2) \cdot \sin t \cdot \cos t / (a^2 \sin^2 t + c^2 \cos^2 t)$

$\Gamma^t_{\theta\theta} = \langle X_{\theta\theta}, X_t \rangle \cdot g^{tt} + \langle X_{\theta\theta}, X_\theta \rangle \cdot g^{\theta t} = a^2 \sin t \cos t / (a^2 \sin^2 t + c^2 \cos^2 t)$

$\Gamma^\theta_{t\theta} = \langle X_{t\theta}, X_t \rangle g^{t\theta} + \langle X_{t\theta}, X_\theta \rangle \cdot g^{\theta\theta} = -a^2 \sin t \cos t / a^2 \cos t = -\tan t.$

And, $\Gamma^t_{t\theta} = \Gamma^\theta_{tt} = \Gamma^\theta_{\theta\theta} = 0.$

「2025. Fall

1. a). Show that $\phi : P_n(\mathbb{R}) \to \mathbb{R} : p(t) \mapsto p'(0)$ is linear functional. $\boxed{8m}$

b). Find a dual basis corresponding to $\{1, t, \cdots, t^n\}$ of $P_n(\mathbb{R})$.

Proof. Let $f(x) = \sum_{i=0}^{n} a_i x^i$ and $g(x) = \sum_{i=0}^{n} b_i x^i$ and $c \in \mathbb{R}$ be given.

Then,
$$\phi(cf+g) = (cf+g)'|_{t=0} = c \cdot f'|_{t=0} + g'|_{t=0} = c \cdot \phi(f) + \phi(g)$$

by the linearity of diffrentiation, This shows the a),

To show b), let

$$\mathcal{B}^* := \left\{ \frac{1}{k!} \cdot \phi_k : P_n(\mathbb{R}) \to \mathbb{R} : p(t) \mapsto p^{(k)}(0) \;\middle|\; k = 0, 1, \cdots, n \right\}$$

Then, for each $k = 0, 1, \cdots, n$,

$$\frac{1}{k!} \cdot \phi_k(t^k) = \frac{k!}{k!} \cdot 1 = 1, \quad \text{and} \quad \frac{1}{k!} \cdot \phi_k(t^r) = 0 \text{ if } r \neq k.$$

Similarly a), the elements in $\mathcal{B}^*$ are all linear functional, So the $\mathcal{B}^*$ is dual basis.

[2025. Fall

2. Let $A \in M_{m\times m}(\mathbb{R})$. $\boxed{15m}$: $A^{\underline{n}}$ 이라 얼굴, $n=2$인때부터 생각

a). For $n \in \mathbb{N}$, $\operatorname{rank} A^n \le \operatorname{rank} A$.

b). Let $L_A : \mathbb{R}^m \to \mathbb{R}^m ; x \mapsto Ax$.

Show that for $n \ge 2$,

$$\operatorname{rank} A^n = \operatorname{rank} A \quad \text{implies} \quad \operatorname{Ker} L_A \cap \operatorname{Im} L_A = \{0\}.$$

Sol) a). Given $x \in \operatorname{Im} A^n$, $x = A^n y$ for some $y \in \mathbb{R}^m$.

So, $x = A^n y = A^{n-1} A y \in \operatorname{Im} A$, for $n \ge 2$. Thus $\operatorname{Im} A^n \subseteq \operatorname{Im} A$, it follows $\operatorname{rank} A^n \le \operatorname{rank} A$. If $n=1$, Clear.

b). Given $x \in \operatorname{Ker} A$, $A^n x = A^{n-1} A x = A^{n-1} 0 = 0$, so $x \in \operatorname{Ker} A^n$, so $\operatorname{Ker} A \subseteq \operatorname{Ker} A^n$.

Since $\operatorname{rank} A^n = \operatorname{rank} A$, the dimension theorem gives $\operatorname{nullity} A^n = \operatorname{nullity} A$.

Therefore, since $\mathbb{R}^m$ is finite dimensional, So is $\operatorname{Ker} A^n$, thus $\operatorname{Ker} A = \operatorname{Ker} A^n$.

Now, let $x \in \operatorname{Ker} L_A \cap \operatorname{Im} L_A$ be given. Then, $Ax = 0$ and $x = Ay$ for some $y \in \mathbb{R}^m$.

Then, $A^{n-1} \cdot x = A^n y = 0$, because $Ax = 0$, so $A^{n-2} \cdot Ax = 0$, and this implies $y \in \operatorname{Ker} A^n$.

Since $\operatorname{Ker} A^n = \operatorname{Ker} A$, $Ay = 0$, so $x = Ay = 0$, so proved. ⊣

「2025. Fall

3. Find all $a,b,c \in \mathbb{R}$ such that $\boxed{6m}$

$$A = \begin{bmatrix} 2 & 0 & 0 \\ a & 2 & 0 \\ b & c & -1 \end{bmatrix}$$ is diagonalizable.

Sol). Since the Charatorstic polynomial of A is regardless of a,b,c,

$$\chi_A(t) = (t-2)^2(t+1).$$

Since a Matrix is diagonalizable if and only if the minimal polynomial splits and separable, and the minimal polynomial divides the charatorstic polynomial, the A is diagonalizable if and only if

$$(A-2I)(A+I) = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & -3 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 & 0 \\ a & 3 & 0 \\ b & c & 0 \end{bmatrix} = 0.$$

Calculating gives

$$= \begin{bmatrix} 0 & 0 & 0 \\ 3a & 0 & 0 \\ ac & 0 & 0 \end{bmatrix}$$

therefore, $a=0$, $b \in \mathbb{R}$, $c \in \mathbb{R}$ are all possible (a,b,c).

「2025. Fall.

20m

4. Let $N \subseteq GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be the set of all pairs (A, B) such that
$$\det A = \pm \det B.$$

Show that N is normal and determine the factor group $(G\times G)/N$.

Sol). Let $(A,B) \in N$ and $(C,D) \in GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be given. Then
$$(C,D)(A,B)(C,D)^{-1} = (CAC^{-1}, DBD^{-1})$$

And, since the determinant preserves the multiplication of matrix,
$$\det CAC^{-1} = \det C \cdot \det C^{-1} \cdot \det A = \det A$$
and $\det DBD^{-1} = \det D \cdot \det D^{-1} \cdot \det B = \det B$

so, since $(A,B) \in N$, $\det CAC^{-1} = \det A = \pm \det B = \pm \det DBD^{-1}$

therefore $(C,D)(A,B)(C,D)^{-1} \in N$. And, to show N is subgroup,
first, $(I, I) \in N$ trivially, so $N \neq \emptyset$. Let $(A,B), (C,D) \in N$ be given. Then
$$(A,B)\cdot(C,D)^{-1} = (AC^{-1}, BD^{-1})$$

Satisfies $\det AC^{-1} = \det A \cdot \det C^{-1} = \pm \det B \cdot \det C^{-1} = \pm \det B \cdot \det D^{-1} = \pm \det BD^{-1}$.
therefore, by the subgroup criterion, N is subgroup, and normal. $\square$

Define a map
$$\varphi : GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) \to \mathbb{R}^{\times} : (A,B) \mapsto (\det A \cdot \det B^{-1})^2$$

Then, $\ker \varphi = N$ and φ is homomorphism, onto $(0, \infty)$, because the det preserves multiplication, so is φ, and ontoness is given by: fix $B = I$, and $A = \mathrm{diag}(\sqrt{r}, 1, 1, \dots, 1)$ where $r > 0$ is arbitrary.

By the first-isomorphism theorem,
$$GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) / N \cong ((0,\infty), \times)$$
isom. φ

Comment: φ 을 먼저 찾고 시작하고, hom 임을 보이면 앞 계산 과정이 날라감!

2025. Fall

10m

5. Let G be an order 21 subgroup of $\mathbb{C}^\times$.

a) G is cyclic.

b). For any $g \in G$, $x^{12} = g^{15}$ has 3 distinct solutions in G.

Proof. a). Since $\mathbb{C}$ is abelian, so G is also abelian.
By the Cauchy's theorem, G has order 3 element x and order 7 element y.
Now, since 3 and 7 are relatively prime, $|xy| = 21$, so $G = \langle xy \rangle$.

b) By a), $G \cong \mathbb{Z}/21\mathbb{Z}$. So, we can write the given equation as

$$12x \equiv 15g \mod 21 \quad (g \in \mathbb{Z}/21\mathbb{Z}).$$

Regardless of g, $\gcd(12, 21) = 3$ divides $15g$, so it has three distinct solutions.

Note: $ax \equiv b \mod n$ has a solution iff $\gcd(a,n) \mid b$.
And in such a case, it has $\gcd(a,n)$ solution

「2025. Fall,

6. Let $\{a_n\}$ be a bounded sequence in $\mathbb{R}$, Define $\boxed{22m}$

$$\{a_n\}^* = \{x \in \mathbb{R} \mid x \text{ is a subsequential limit of } \{a_n\}\}$$

a). Prove that $\{a_n\}^*$ is closed.

b). Show $\limsup_{n\to\infty} a_n = \sup\{a_n\}^*$

c). Is there a sequence $\{b_n\}$ such that $\{b_n\}^* = [0,1]$?

Proof a). If $\{a_n\}^*$ is finite, then clearly closed in $\mathbb{R}$.
Assume that $\{a_n\}^*$ is infinite. Since $\{a_n\}$ is bounded, so the set of subsequential limits $\{a_n\}^*$ is bounded set. So it has a limit point.
Let a be a limit point of $\{a_n\}^*$. Then, given $k \in \mathbb{N}$, there exists a subsequential limit p of $\{a_n\}$ such that $p \in B_{\frac{1}{k}}(a) \setminus \{a\}$.
For $k=1$, choose such a $p_1 \in B_1(a) \setminus \{a\}$, and since p is a subsequential limit, choose $a_{p_1} \in B_{1-|p-a|}(p)$, then $a_{p_1} \in B_1(a)$.

For chosen $a_{p_1}, \dots, a_{p_{k-1}}$, choose $p_k \in B_{\frac{1}{k}}(a) \setminus \{a\}$,
Then, similarly, there is $a_{p_k} \in B_{\frac{1}{k}}(a)$ such that $p_k > p_{k-1}$,

because no such p_k exists, then it is contradiction to a_{p_k} is a subsequential limit. Therefore, the arbitrary limit point a is contained in $\{a_n\}^*$,
So it is closed. ⌟

b). Since a), $\{a_n\}^*$ is closed and bounded, so compact. Thus

$$\sup\{a_n\}^* = \max\{a_n\}^* = a \text{ for some subsequential limit } a.$$

So, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N \Rightarrow a_n < a + \varepsilon$ (since $a = \max\{a_n\}^*$)
and there exists infinitely many terms in $\{a_n\}$ such that $a - \varepsilon < a_n$ (since $a_{n_k} \to a$)
Thus $\limsup_{n\to\infty} a_n = a$.

c). Let b_n be defined as:

$$\frac{1}{1}, \frac{1}{2}, \frac{2}{2}, \frac{1}{3}, \frac{2}{3}, \frac{3}{3}, \dots$$

Then, given $b \in [0,1]$, there exists a subsequence in $\{b_n\}$ converges to b, because: for any $\varepsilon > 0$, there are infinitely many terms in b_n such that

$$|b_n - b| < \varepsilon,$$

So we can construct the subsequence. ☒

「2025, Fall.

17. Let E be a nonempty bounded subset of a Metric space (X,d). 10m

Define $\quad \operatorname{diam} E := \sup\{d(x,y) \mid x,y \in E\}$

a). Prove $\operatorname{diam} E = \operatorname{diam} \overline{E}$

b). If E is Compact, then $d(x',y') = \operatorname{diam} E$ for some $x',y' \in E$.

Proof.

a). Since $E \subseteq \overline{E}$, $\operatorname{diam} E \leq \operatorname{diam} \overline{E}$ is clear.

To show the inverse inequality, Using the Property of Supremum,

Given $\varepsilon > 0$, there exists $x,y \in \overline{E}$ such that

$$\operatorname{diam} \overline{E} - \frac{\varepsilon}{2} < d(x,y) \leq \operatorname{diam} \overline{E}$$

And, since $\overline{E}$ is closure, there are $x',y' \in E$ such that

$$d(x',x) < \frac{\varepsilon}{4} \quad \text{and} \quad d(y,y') < \frac{\varepsilon}{4}.$$

Now, $\operatorname{diam} \overline{E} < d(x,y) + \frac{\varepsilon}{2} < d(x,x') + d(x',y') + d(y',y) + \frac{\varepsilon}{2} < d(x',y') + \varepsilon$.

Since $\varepsilon > 0$ was arbitrary, $\operatorname{diam} \overline{E} \leq \operatorname{diam} E$, So Proved. 7m.

b), Since E is Compact set, so $E \times E$ is also Compact set.

Since the metric function d is continuous, it Preserves Compactness,

so $d[E \times E] \subseteq \mathbb{R}$ is Compact set. Therefore,

$$\operatorname{diam} E = \max d[E \times E] \in d[E \times E],$$

So there is $(x',y') \in E \times E$ such that $d(x',y') = \max d[E \times E]$ 3m.

「2025. Fall.

8. Let $a>1$. Evaluate $\boxed{5m}$

$$\lim_{t\to0^+}\int_t^{at}\frac{\cos x}{x}\,dx$$

pruf. Consider the taylor expansion

$$\cos x=\sum_{n=0}^{\infty}(-1)^n\cdot\frac{x^{2n}}{(2n)!}$$

Therefore $\dfrac{\cos x}{x}=\dfrac{1}{x}-\dfrac{x}{2!}+\dfrac{x^3}{4!}-\cdots$

Now, $$\int_t^{at}\frac{\cos x}{x}\,dx=\int_t^{at}\frac{1}{x}+\int_t^{at}-\frac{x}{2!}+\frac{x^3}{4!}-\cdots\,dx$$

$$=\ln a+\sum_{n=1}^{\infty}\int_t^{at}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}$$

(It is possible, because $\sum(-1)^n\cdot x^{2n-1}/(2n)!$ converges for all $x\in\mathbb{R}$)

Now, $\displaystyle\lim_{t\to0^+}\sum_{n=1}^{\infty}\int_t^{at}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}\,dx=0$, so

the given limit is $\ln a$.

Revised in next.

「Observation for $\lim_{t\to 0^+} \int_t^{at} \frac{\cos x}{x}\, dx$.

As we know, the power series representation

$$\frac{1}{x}\cdot \cos x = \frac{1}{x}\cdot \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = \frac{1}{x} + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!}$$

is valid for all $x \neq 0$. Therefore, for ant $t>0$,

$$\int_t^{at} \frac{\cos x}{x}\, dx = \int_t^{at} \frac{1}{x} + \sum_{n=1}^{\infty} (-1)^n \cdot \frac{x^{2n-1}}{(2n)!}\, dx$$

$$= \int_t^{at} \frac{1}{x}\, dx + \int_t^{at} \sum_{n=1}^{\infty} (-1)^n \cdot \frac{x^{2n-1}}{(2n)!}\, dx$$

Meanwhile, the series converges uniformly in $[t, at]$ for small enough $t>0$, specifically, $t < \frac{1}{a}$, then for each $n \in \mathbb{N}$,

$$\left| (-1)^n \cdot \frac{x^{2n-1}}{(2n)!} \right| \leq \left| \frac{1}{(2n)!} \right| \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{(2n)!} < \infty,$$

$\frac{1}{2n} x^{2n}$

therefore the Weierstrass M-test gives the condition,

Thus, we can exchange the integral to the sum,

$$\int_t^{at} \sum_{n=1}^{\infty} (-1)^n \cdot \frac{x^{2n-1}}{(2n)!} = \sum_{n=1}^{\infty} \int_t^{at} (-1)^n \cdot \frac{x^{2n-1}}{(2n)!} = \sum_{n=1}^{\infty} (-1)^n \cdot \left(\frac{1}{2n\cdot(2n)!} \cdot \left(a^{2n} t^{2n} - t^{2n}\right)\right)$$

Converges uniformly similarly, now, we can exchange the limit to the sum

$$\lim_{t\to 0^+} \sum_{n=1}^{\infty} (-1)^n \cdot \frac{1}{2n\cdot(2n)!} \cdot t^{2n}(a^{2n}-1)$$

$$= \sum_{n=1}^{\infty} \lim_{t\to 0^+} \quad //$$

$$= \sum_{n=1}^{\infty} 0$$

$$= 0$$

Consequently, the given integral is

$$\int_t^{at} \frac{1}{x}\, dx = \ln at - \ln t = \ln a.$$

Note: $\frac{\cos x}{x} = \frac{1}{x} + \frac{\cos x - 1}{x}$, $\left| \int_t^{at} \frac{\cos x - 1}{x}\, dx \right| \leq \int_t^{at} \frac{x}{2} \to 0$ as $t \to 0$.

「2025. Fall.

12. Define an equi. rel on $\mathbb{R}$ as $x \sim y \iff x-y \in \mathbb{Q}$.
 " as $x \approx y \iff x=y$ or $x,y \in \mathbb{Q}$.

a). Describe the quotient topology on $\mathbb{R}/\sim$.

b). " on $\mathbb{R}/\approx$.

c). In $\mathbb{R}/\approx$, determine if every point is closed

d). " the image of the set of irrationals under quotient map is compact.

Proof. Since $\mathbb{Q}$ is closed under the operation $-$, so these relations are equi. rel.

a). First, $\mathbb{R}/\sim = \{a+\mathbb{Q} \mid a \in \mathbb{R}\}$, where $a+\mathbb{Q} = \{a+q \mid q \in \mathbb{Q}\}$. And, given a basis element $(a,b) \subseteq \mathbb{R}$, for the projection $\pi : \mathbb{R} \to \mathbb{R}/\sim : a \mapsto a+\mathbb{Q}$.

$$\pi^{-1}[\pi[(a,b)]] = \bigcup_{r \in (a,b)} r+\mathbb{Q} = \bigcup_{q \in \mathbb{Q}} (a+q, b+q)$$

is open and since π is quotient map, so $\pi[(a,b)]$ is open, so π is open.
Therefore, the topology on $\mathbb{R}/\sim$ is generated by

$$\mathcal{B}_\sim = \{\pi[(a,b)] \mid a<b\} = \{(a,b)+\mathbb{Q} \mid a<b\}.$$

But, by the density of rationals, for any $a<b$, $(a,b)+\mathbb{Q} = \mathbb{R}/\mathbb{Q}$.
Therefore the topology on $\mathbb{R}/\sim$ is indiscrete.

⌈2025. Fall.

13. Consider a regular plane curve $C: I \to \mathbb{R}^2$.

a). Give a formula for the normal n to C, and define the signed curvature.

b). Explain the relation between the total angle change and the signed curvature.

c). Assume C is a closed plane curve parametrized by arc-length, and $I = [0, l]$.
Prove or disprove that: there are $t, t' \in I$ such that $C'(t) = -C'(t')$.
and if $t = 0$, there is such a $t' \in I$?
Moreover, show that $\int_0^l |k(s)| ds \geq 2\pi$ where $k(s)$ is the signed curvature.

Proof. Denote $C(t) = (x(t), y(t))$.

a). The tangent vector is

$$T(t) = \frac{C'(t)}{|C'(t)|} = \frac{(x'(t), y'(t))}{\sqrt{x'(t)^2 + y'(t)^2}}$$

So, the normal vector is

$$N(t) = \frac{(-y'(t), x'(t))}{\sqrt{x'(t)^2 + y'(t)^2}}$$

And, since the curvature $k(s(t))$ satisfies $T'(t) = k(t) \cdot N(t)$,

$$k(t) = k(t) \langle N(t), N(t) \rangle = \langle T'(t), N(t) \rangle$$

⌈Debris.

Denote $C = (x, y)$. Then,

$$T = (x', y') / |(x', y')| = \frac{(x', y')}{\sqrt{x'^2 + y'^2}}$$

is the tangent vector, and the normal vector is

$N = \frac{(-y', x')}{\sqrt{x'^2 + y'^2}}$. Moreover, the signed curvature k satisfies $kN = T'$,

so $k = k \cdot \langle N, N \rangle = \langle T', N \rangle \cdot |C'(t)|^{-1}$ Meanwhile

$$T' = \frac{(x'', y'')}{\sqrt{x'^2 + y'^2}} + (x', y') \cdot \left(-\frac{1}{2} \cdot \frac{2x'x'' + 2y'y''}{(x'^2 + y'^2)^{\frac{3}{2}}}\right) \text{ so}$$

$$k = \frac{1}{x'^2 + y'^2} \cdot (x'y'' - x''y') \cdot \frac{1}{\sqrt{x'^2 + y'^2}}$$

TSNU. Qual.

Prove that $\lim_{b\to\infty}\int_0^b \frac{\sin x}{x}\,dx$ exists.

Proof. First, let $a_n = n\pi$ $(n\in\mathbb{N})$. Then, a sequence $\left\{\int_0^{a_n}\frac{\sin x}{x}\,dx\right\}$ converges,
because: $\left|\int_{n\pi}^{(n+1)\pi}\frac{\sin x}{x}\,dx\right|$ is decreasing and converges to zero, and

$\int_{n\pi}^{(n+1)\pi}\frac{\sin x}{x}\,dx$ is positive if n is even and
is negative if n is odd.

That is, $\int_0^{n\pi}\frac{\sin x}{x}\,dx = \sum_{k=1}^{n}\int_{a_{k-1}}^{a_k}\frac{\sin x}{x}\,dx$ is alternating series, so converges.

「2026. Spring.

$\boxed{2nm}$

3. Let V be an n-dimensional vector space over $\mathbb{C}$,
Let $T: V \to V$ be a linear operator such that $T^3 = T$.
a). Find all possible characteristic polynomials and corresponding minimal polynomial for T.
b). Prove $V = \ker T \oplus \operatorname{Im} T$.

Proof.

a). Since $T^3 - T = 0$, the minimal polynomial $m(t)$ for T divides $t^3 - t = 0$.
$t^3 - t = t(t-1)(t+1)$, there are possibilities:

$m(t) = t$ or $t-1$ or $t+1$ or $t(t-1)$ or $t(t+1)$ or $(t-1)(t+1)$ or $t(t-1)(t+1)$.

Since $m(t)$ divides the characteristic polynomial and they have same characteristic values, the possible characteristic polynomial is:

$$t^{r_1} \cdot (t-1)^{r_2} \cdot (t+1)^{r_3}$$ where r_1, r_2, r_3 are non-negative integers and $r_1 + r_2 + r_3 = n$.

If $r_1 = 0$ and $r_2, r_3 \neq 0$, then the corresponding minimal polynomial is
$$(t-1)(t+1)$$
and other cases are similar.

b). ~~Given $\alpha \in \ker T \cap \operatorname{Im} T$, $T\alpha = 0$ and $\alpha = T\beta$ for some $\beta \in V$.~~

Given $\alpha \in V$, $\alpha = T^2\alpha + \alpha - T^2\alpha$ and since $T(\alpha - T^2\alpha) = (T - T^3)\alpha = 0\alpha = 0$,
so $\alpha - T\alpha^2 \in \ker T$ and trivially $T^2\alpha \in \operatorname{Im} T$. Moreover, if $T\alpha = 0$ and $\alpha \in \operatorname{Im} T$,
Then $\alpha = T\beta = T^3\beta$ for some $\beta \in V$ and $T\alpha = T^2\beta = 0$ so $T^3\beta = T\,T^2\beta = \alpha = 0$.
so $V = \ker T \oplus \operatorname{Im} T$. ∎

[Modified.]

Let T be a linear operator on the fin. dim space V over $\mathbb{C}$ satisfying $T^4 = T^2$.

a). Prove that $V = \ker T^2 \oplus \operatorname{Im} T^2$.

b). Describe $T|_{\operatorname{Im} T^2}$.

c). Prove that T is diagonalizable $\Leftrightarrow$ $\ker T = \ker T^2$.

observe: $T^4 - T^2 = 0$, so the minimal polynomial divides $t^2(t-1)(t+1)$.

a). Given $\alpha \in V$, $\alpha = T^2\alpha + \alpha - T^2\alpha$ and $T^2\alpha \in \operatorname{Im} T^2$ clearly, and $T^2(I - T^2)\alpha = (T^2 - T^4)\alpha = 0 \cdot \alpha = 0$, so $\alpha - T^2\alpha$, which proves $\alpha - T^2\alpha \in \ker T^2$, so $V = \ker T^2 + \operatorname{Im} T^2$. Moreover, given $\alpha \in \ker T^2 \cap \operatorname{Im} T^2$, $T^2\alpha = 0$ and $\alpha = T^2\beta$ for some $\beta \in V$, so

$$\alpha = T^2\beta = T^4\beta = T^2 T^2 \beta = T^2\alpha = 0$$

So $V = \ker T^2 \oplus \operatorname{Im} T^2$.

b). Given $\alpha \in \operatorname{Im} T^2$, $\alpha = T^2\beta$ for some $\beta \in V$. So, $T\alpha = T^3\beta$. Multiplying T, then $T^2\alpha = T^4\beta = T^2\beta$, so $T^2(\alpha - \beta) = 0$. Therefore, $\alpha - \beta \in \ker T^2$, but since $\alpha \in \operatorname{Im} T^2$ and $\operatorname{Im} T^2 \cap \ker T^2 = 0$, $\beta = \alpha + \gamma$, where $\gamma \in \ker T^2$. Now, $T\alpha = T^3\beta = T^3\alpha + T^3\gamma = T^3\alpha$, So, $T|_{\operatorname{Im} T^2} = T^3|_{\operatorname{Im} T^2}$. moreover, since $T^2 = T^4$, $T^2|_{\operatorname{Im} T^2} = I$.

c). Since T is diagonalizable $\Leftrightarrow$ minimal polynomial of T splits and separable. That is, the minimal polynomial is $t^a(t-1)^b(t+1)^c$, $a, b, c \in \{0, 1\}$.

2026 Spring

5. Let $f(x) = x^4 + 2x + 3 \in \mathbb{F}_7[x]$ and $K = \mathbb{F}_7[x]/(f)$.
Let $\varphi : K \to K : a \mapsto a^7$.

a). Prove K is a field and φ is Automorphism.

b). Let $G = \langle \varphi \rangle \le \mathrm{Aut}(K)$. Determine the number of $a \in K$ such that $|\mathrm{Orb}(a)| = 2$.

Proof

a). Since $f(0) = 3$, $f(1) = 6$, $f(2) = 2$, $f(3) = 6$, $f(4) = 1$, $f(5) = 1$, $f(6) = 2$,
So $f(x)$ has no linear factor. So, if $f(x)$ is reducible,
$$f(x) = (x^2 + ax + b)(x^2 + cx + d) \text{ for some } a, b, c, d \in \mathbb{F}_7.$$

Then, $\begin{cases} a + c = 0 \\ ac + b + d = 0 \\ ad + bc = 2 \\ bd = 3 \end{cases}$. If $a = 0$ or $c = 0$, then $a = c = 0$, so $0 = 2$.
If $b = 0$ or $d = 0$, then $0 = 3$,
Therefore, a, b, c, d are non-zeros.
Reduce the system;
$$\Rightarrow \begin{cases} b + d = a^2 \\ a(d - b) = 2 \\ bd = 3 \end{cases}$$
If $b = d$, then $b^2 = 3$, but $\sqrt{3} \notin \mathbb{F}_7$, so $b \ne d$.
And, $3 = 1 \cdot 3$ or $2 \cdot 5$ or $4 \cdot 6$, so
$a^2 = 4$ or 0 or 3. Since $\sqrt{3} \notin \mathbb{F}_7$ and $a \ne 0$, $a^2 = 4$, so $a = 2$ or 5,
with $\{b, d\} = \{1, 3\}$. But,
$$2 \cdot (1 - 3) = 3 \ne 2,$$
$$2 \cdot (3 - 1) = 4 \ne 2,$$
$$5 \cdot (1 - 3) = 4 \ne 2$$
$$5 \cdot (3 - 1) = 3 \ne 2$$
So contradiction. Therefore, f must be an irreducible.
Since $\mathbb{F}_7$ is a field, $\mathbb{F}_7[x]$ is P.I.D., so (f) is maximal, so $\mathbb{F}_7[x]/(f)$ is a field. And, since the characteristic of K is 7, for any $a, b \in K$,
$$(a + b)^7 = \sum_{k=0}^{7} \binom{7}{7-k} a^{7-k} b^k = a^7 + b^7$$
and $(ab)^7 = a^7 b^7$, so φ is Homomorphism. Moreover,
$\varphi(a) = 0 \iff a^7 = 0 \iff a = 0$, so $\ker \varphi$ is trivial, so φ is 1-1.
Now, since $[K : \mathbb{F}_7] = \deg f = 4$, so $\# K = 7^4$, finite, so φ is bijective.

b). Since every finite field is perfect, so $\mathbb{K}$ is Galois over $\mathbb{F}_7$,
So, $|Aut(\mathbb{K})| = \deg f = 4$,
Consider an Action

$$G \times \mathbb{K} \to \mathbb{K} : (\varphi^n, a) \mapsto \varphi^n(a).$$

So. $|Orb(a)| = 2 \iff \varphi^2(a) = a$ and $a \neq 0 \iff a^{49} = a$ and $a \neq 0$
So, the fixed field by $\langle \varphi^2 \rangle$, $\mathbb{K}^{\langle \varphi^2 \rangle} \setminus \{0\}$ is we desired.

Revised in the next

⌈Revised.

Prove that $f(x) = x^4 + 2x + 3$ is irreducible over $\mathbb{F}_7$.
And describe the Galois group of $f(x)$.

1). $f(x)$ has no linear factor.
$f(0) = 3 \neq 0$. And, since $\mathbb{F}_7^\times$ is a group of order 6 under the multiplication.
Therefore, for any $a \in \mathbb{F}_7^\times$, $a^6 = (a^3)^2 = 1$ implies $a^3 = \pm 1$.
If $a^3 = 1$, then $a^4 = a$, so $f(a) = a^4 + 2a + 3 = 3(a+1) = 0$ implies $a = -1$,
but $(-1)^3 = -1 = 6 \neq 1$, so $a^3 \neq 1$.
If $a^3 = -1$, then $a^4 = -a$, so $f(a) = a + 3 = 0$ implies $a = 4$,
but $4^3 = 64 = 1 \neq -1$, so $a^3 \neq -1$. Now, $f(x)$ has no root in $\mathbb{F}_7^\times$.

2). $f(x)$ has no factor of degree 2.
Suppose that $f(x) = (x^2 + ax + b)(x^2 + cx + d)$. Then, we have a system:

$$\begin{cases} a + c = 0 \\ ac + b + d = 0 \\ ad + bc = 2 \\ bd = 3. \end{cases} \Rightarrow \begin{cases} b + d = a^2 \\ a(d - b) = 2 \\ bd = 3 \end{cases} \Rightarrow \begin{cases} (d-b)^2 = (b+d)^2 - 4bd = a^4 - 12 = a^4 + 2 \\ a^2(d-b)^2 = 4 \end{cases}$$

$a \neq 0$, so $a^6 = 1$

$\Rightarrow a^6 + 2a^2 = 1 + 2a^2 = 4 \Rightarrow a^2 = 5$. But, $\{x^2 \mid x \in \mathbb{F}_7\} = \{0, 1, 4, 2\}$,
so $\sqrt{a} \notin \mathbb{F}_7$.

Now, f is irreducible $\Rightarrow \mathbb{F}_7[x]/(f)$ is a field.

「2026 Spring

8. Let $\mathbb{D}$ be the unit open disc in $\mathbb{C}$. Prove or disprove:

Every holomorphic bijection $f: \mathbb{D} \to \mathbb{D}$ has a point $w \in \mathbb{D}$ such that $f(w) = w$.

proof. Let $f: \mathbb{D} \to \mathbb{D}$ be a holomorphic bijection

Suppose that for all $w \in \mathbb{D}$, $f(w) \neq w$. Then, a map

$$z \mapsto \frac{1}{f(z) - z} \quad (z \in \mathbb{D})$$ is well-defined,

[2026 Fall.

2. Let $A \in M_{4\times4}(\mathbb{R})$ be a symmetric with eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4$.

Let
$$P = \frac{1}{2} \cdot \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix},$$

a) Prove that P is an orthogonal projection.

b). Prove $tr(AP) \leq \lambda_1 + \lambda_2$

c) $\displaystyle\max_{\substack{x,y \in \mathbb{R}^4 \\ \|x\|=\|y\|=1 \\ \langle x,y\rangle = 0}} x^T A x + y^T A y = \lambda_1 + \lambda_2.$

a). By just calculation, $P^2 = P = P^T$, and in $\mathbb{R}$, $P^T = P^*$, so P is an orthogonal projection, rank $P = 2$, because the row vectors space has dimension 2.

b). Since P is orthogonal projection, $\mathbb{R}^4 = \operatorname{Im} P \oplus \ker P$, with $(\ker P)^\perp = \operatorname{Im} P$, and rank P = nullity $P = 2$. Since A is symmetric, there are four eigenvectors $\{\alpha_1, \cdots, \alpha_4\}$ corresponding to $A\alpha_i = \lambda_i \alpha_i$ $(i = 1, \cdots, 4)$ and $\{\alpha_1, \cdots, \alpha_4\}$ is orthonormal.

And, let $\{v_1, v_2\}$ be a basis for $\operatorname{Im} P$ and $\{v_3, v_4\}$ be a basis for $\ker P$, satisfying $\{v_1, \cdots, v_4\}$ is orthonormal.

Since the trace is invariant under the similar, for $B = (v_1 \ \cdots \ v_4)$,

$$tr(AP) = tr(B^T A P B) = v_1^T A P v_1 + \cdots + v_4^T A P v_4 = v_1^T A v_1 + v_2^T A v_2$$
$$= \langle Av_1, v_1\rangle + \langle Av_2, v_2\rangle$$

Now, since $v_1, v_2 \in \langle \alpha_1, \cdots, \alpha_4\rangle$ and $\|v_1\| = \|v_2\| = 1$ and $\langle v_1, v_2\rangle = 0$, $v_1 = a_1\alpha_1 + \cdots + a_4\alpha_4$ and $v_2 = b_1\alpha_1 + \cdots + b_4\alpha_4$ such that $a_1^2 + \cdots + a_4^2 = b_1^2 + \cdots + b_4^2 = 1$ and $a_1b_1 + \cdots + a_4b_4 = 0$. Now,

$$\langle Av_1, v_1\rangle + \langle Av_2, v_2\rangle = a_1^2\lambda_1 + \cdots + a_4^2\lambda_4 + b_1^2\lambda_1 + \cdots + b_4^2\lambda_4$$
$$= \lambda_1(a_1^2 + b_1^2) + \cdots + \lambda_4(a_4^2 + b_4^2)$$

But, for each $1 \leq i \leq 4$, $a_i^2 + b_i^2 \leq 1$ and $a_1^2 + \cdots + a_4^2 + b_1^2 + \cdots + b_4^2 = 2$, therefore

$$\leq \lambda_1 \cdot 1 + \lambda_2 \cdot 1 = \lambda_1 + \lambda_2.$$ It proves b)

c) Given $x, y \in \mathbb{R}^4$ such that $\langle x, y\rangle = 0$, with $\|x\| = \|y\| = 1$, $x^T A x + y^T A y = \langle Ax, x\rangle + \langle Ay, y\rangle \leq \lambda_1 + \lambda_2$, by the argument in b).

And, take $x = \alpha_1$ and $y = \alpha_2$, we obtain c).

[2026 Fall.

4. Let G be a group and $H_1, H_2 \leq G$ be subgroups of finite index.

a). Prove that the map

$$T : G \to S_{G/H_1 \times G/H_2} : x \mapsto T_x \text{ where } T_x(aH_1, bH_2) = (xaH_1, xbH_2)$$

is Homomorphism and $\operatorname{Ker} T \subseteq H_1 \cap H_2$.

b). Prove $H_1 \cap H_2$ has finite index.

Proof

a). Given $g_1, g_2 \in G$, $T(g_1 g_2)(aH_1, bH_2) = (g_1 g_2 aH_1, g_1 g_2 bH_2)$

$$= T(g_1) \circ T(g_2)(aH_1, bH_2)$$

for each $a, b \in G$, so it preserves the operation.

And $g \in \operatorname{Ker} T \iff T_g = id \iff$ for all $a, b \in G$, $T_g(aH_1, bH_2) = (gaH_1, gbH_2) = (aH_1, bH_2)$

$\iff$ for all $a, b \in G$, $a^{-1}ga \in H_1$ and $b^{-1}gb \in H_2$

$\Rightarrow g \in H_1 \cap H_2$ (take $a = b = 1_G$)

b). By the first-isomorphism theorem,

$$G/\operatorname{Ker} T \cong \operatorname{Im} T \leq S_{G/H_1 \times G/H_2}$$

and since H_1, H_2 have finite index, so the Codomain of T is finite set.

Now, $|G : H_1 \cap H_2| \leq |G : \operatorname{Ker} T| \leq \# S_{G/H_1 \times G/H_2}$

because $\operatorname{Ker} T \subseteq H_1 \cap H_2$, so proved.

Note: If $A \leq B \leq G$ and $|G : A|$ is finite, then

$$G/A \to G/B : gA \mapsto gB$$

is well-defined and onto, because: Since $A \subseteq B$,

$$g_1 A = g_2 A \iff g_1 g_2^{-1} \in A \leq B \Rightarrow g_1 g_2^{-1} \in B \iff g_1 B = g_2 B$$

and surjectiveness is clear.

[2026. Fall.

8. For $f \in C([0,1])$, define its extension

$$\tilde{f}(x) := \begin{cases} f(0) & \text{if } x<0 \\ f(x) & \text{if } 0 \le x \le 1 \\ f(1) & \text{if } x>1 \end{cases}$$

For each $n \ge 1$, define

$$P_n(x) := \frac{n}{2} \cdot \chi_{[-\frac{1}{n}, \frac{1}{n}]}(x)$$

Define $T_n f(x) := \int_{\mathbb{R}} P_n(x-y)\tilde{f}(y)\, dy, \quad x \in \mathbb{R}.$

a). Show that $T_n f \to f$ on $[0,1]$,

b). Determine $T_n f \to f$ uniformly on $[0,1]$ or not.

Proof. Given $x \in \mathbb{R}$ and $n \in \mathbb{N}$, $x - y \in [-\frac{1}{n}, \frac{1}{n}] \Leftrightarrow -\frac{1}{n} + x \le y \le \frac{1}{n} + x$. Therefore,

$$T_n f(x) = \int_{-\frac{1}{n}+x}^{\frac{1}{n}+x} \frac{n}{2} \cdot \tilde{f}(y)\, dy.$$

Since $\tilde{f}$ is continuous on $[-1,2]$, so it is continuously uniformly.

Given $\varepsilon > 0$, take $\delta > 0$ such that $\forall x, t \in [0,1]$, $|x-t| < \delta \Rightarrow |f(x) - f(t)| < \varepsilon$.

Now, given $x \in [0,1]$, and $n > \frac{1}{\delta}$,

$$\left| T_n f(x) - f(x) \right| = \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} \tilde{f}(y)\, dy - \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} \cdot f(x)\, dy \right|$$

$$= \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} \cdot [\tilde{f}(y) - f(x)]\, dy \right|$$

$$\le \frac{n}{2} \cdot \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} |\tilde{f}(y) - f(x)|\, dy$$

$$< \frac{n}{2} \cdot \frac{2}{n} \cdot \varepsilon = \varepsilon$$

Therefore, $\sup_{[0,1]} |T_n f - f| \to 0$ as $n \to \infty$, so converges uniformly.

[2026 Fall.

9. Let $f \in C([0,\pi])$ and define

$$(Tf)(x) := \int_0^\pi \min\{x,y\} \cdot f(y)\, dy, \qquad 0 \le x \le \pi$$

a). Show that Tf is twice differentiable on $(0,\pi)$ and satisfies

$$(Tf)'(x) = \int_x^\pi f(y)\, dy, \qquad (Tf)''(x) = -f(x)$$

b). Suppose that $f \ne 0$ and $\lambda \ne 0$ satisfy $Tf = \lambda f$.
Show that $\lambda > 0$ and that f satisfies

$$-f''(x) = \frac{1}{\lambda} \cdot f(x), \qquad 0 < x < \pi$$

With the boundary conditions $f(0)=0$, $f'(\pi)=0$.

c) Determine all $\lambda > 0$ and all non-trivial functions f satisfying the differential equation and the boundary conditions in (b).

Proof. Observe that; given $0 \le x \le \pi$,

$$\int_0^\pi \min\{x,y\} \cdot f(y)\, dy = \int_0^x \min\{x,y\} \cdot f(y)\, dy + \int_x^\pi \min\{x,y\} \cdot f(y)\, dy$$

$$= \int_0^x y f(y)\, dy + x \cdot \int_x^\pi f(y)\, dy$$

a). By the Fundamental Theorem of Calculus and f is Continuous, both $\int_0^x y f(y)\, dx$ and $x \cdot \int_x^\pi f(y)\, dy$ are differentiable, So

$$(Tf)'(x) = x f(x) + \int_x^\pi f(y)\, dy - x f(x) = \int_x^\pi f(y)\, dy$$

And, FTC again, $(Tf)'$ is differentiable, and $(Tf)''(x) = -f(x)$.

b). To show $\lambda > 0$, observe that

$(\lambda f)'' = (Tf)'' = -f(x)$, so $-f'' = \frac{1}{\lambda} f$, $x \in (0,\pi)$.

Moreover, Since the observation above, $(Tf)(0) = 0$, so $\lambda f(0) = 0$, so $f(0) = 0$.

And, Since the fomula in a),

$$f'(\pi) = \frac{1}{\lambda}(Tf)'(\pi) = \frac{1}{\lambda} \cdot \left(\int_\pi^\pi \cdot \right) = 0.$$

By the result of a),

「2026 Fall

9, Let $f \in C([0,\pi])$ and define

$$(Tf)(x) := \int_0^{\pi} \min\{x,y\} \cdot f(y)\, dy, \quad 0 \le x \le \pi$$

a). Show that Tf is twice differentiable on $(0,\pi)$ and satisfies

$$(Tf)'(x) = \int_x^{\pi} f(y)\, dy, \quad (Tf)''(x) = -f(x)$$

b). Suppose that $f \neq 0$ and $\lambda \neq 0$ satisfy $Tf = \lambda f$.
Show that $\lambda > 0$ and that f satisfies

$$-f''(x) = \frac{1}{\lambda} \cdot f(x), \quad 0 < x < \pi$$

With the boundary conditions $f(0)=0$. $f'(\pi)=0$.

c) Determine all $\lambda > 0$ and all non-trivial functions f satisfying the differential equation and the boundary conditions in (b).

Proof of c).

[Another problems. 26/09/08

1. Let $T \in End(V)$ where V is fin. dim v.s over F with $ch(F) \neq 2$.

a) Prove that $V = \ker T \oplus Im\, T$.

b) Prove that $T|_{Im T} : Im\, T \to Im\, T : \alpha \mapsto T(\alpha)$ is an Automorphism.

c) In addition, if $T^3 = T$, prove T is diagonalizable and find all possible eigenvalue.

Proof.

a). Given $\alpha \in \ker T$, $T^2(\alpha) = T(T(\alpha)) = T(0) = 0$, so $\ker T \subseteq \ker T^2$.

Meanwhile, by the dimension theorem and by the assumption,

$$\dim V = \operatorname{rank} T + \operatorname{nullity} T = \operatorname{rank} T^2 + \operatorname{nullity} T = \operatorname{rank} T^2 + \operatorname{nullity} T^2$$

So, $\dim \ker T = \dim \ker T^2$, and $\dim \ker T \leq \dim V < \infty$ gives $\ker T = \ker T^2$. Let $\alpha \in \ker T \cap Im\, T$. Then, $\alpha = T(\beta)$ for some $\beta \in V$ and $T(\alpha) = T^2(\beta) = 0$. So, $\beta \in \ker T^2 = \ker T$, it gives $\alpha = T(\beta) = 0$.

And, let $\{\alpha_1, \cdots, \alpha_k\}$ and $\{\alpha_{k+1}, \cdots, \alpha_n\}$ be the basis for $\ker T$ and $Im\, T$. Then, $\{\alpha_1, \cdots, \alpha_n\}$ is linearly independent, because $\ker T \cap Im\, T = \{0\}$, which proves $V = \ker T \oplus Im\, T$.

b) Since $Im\, T \supseteq Im\, T^2$ and $\operatorname{rank} T = \operatorname{rank} T^2$, $Im\, T = Im\, T^2$ and so

$$T[Im\, T] \subseteq Im\, T$$

And, since $V = Im\, T \oplus \ker T$, $\ker T|_{Im T} = \{0\}$, so the restriction is injective, and also surjective by $\dim V < \infty$. So it is automorphism.

C). $T^3 = T$ means $x(x^2-1)$ is a multiple of the minimal polynomial, but $x(x-1)(x+1)$ is separable and splits over F, so the minimal polynomial also splits and is separable, thus T is diagonalizable. And the possible eigenvalues are $0, \pm 1$.

Note: Given $T : V \to V$, (not necessary finite-dimensional)

$$V = \ker T \oplus Im\, T \quad \text{if and only if} \quad T|_{Im T} \text{ is an Automorphism.}$$

In such the case, given $\alpha \in V$,

$$\alpha = \underbrace{\left[\alpha - (T|_{Im T})^{-1}(T(\alpha))\right]}_{\ker T} + \underbrace{(T|_{Im T})^{-1}(T(\alpha))}_{Im\, T}.$$

2. Let V, W be fin. dim real inn. prod space, and let $T: V \to W$ be injection.
a). Prove that $T^* \circ T: V \to V$ is invertible.
b). Prove that $T \circ (T^* \circ T)^{-1} \circ T^*: W \to W$ is an orthogonal proj of W onto $\operatorname{Im} T$.
Proof.
a). T is 1-1 $\iff \ker T = \{0\}$, since $\dim V < \infty$.
Meanwhile, observe that given $\alpha \in V$,

$$T^*T(\alpha) = 0 \Rightarrow \langle T^*T(\alpha), \alpha \rangle = 0$$
$$\Rightarrow \langle T(\alpha), T(\alpha) \rangle = 0$$
$$\Rightarrow T(\alpha) = 0$$

So $\ker T^*T \subset \ker T = \{0\}$, thus $\ker T^*T = \{0\}$.
Since $\dim V < \infty$, T^*T is 1-1 and onto.

b) observe that

$$P^2 = T(T^*T)^{-1}T^*T(T^*T)^{-1}T^* = T(T^*T)^{-1}T^* = P$$

$P^* = P$, since $(ABC)^* = (BC)^*A^* = C^*B^*A^*$ and $(T^{-1})^* = (T^*)^{-1}$.
So, $P^2 = P^* = P$, P is orthogonal projection.
And, since T is 1-1, so is T^*, thus T^* is onto, so

$$\operatorname{Im} P = T(T^*T)^{-1}T^*[W] = T[W] = \operatorname{Im} T.$$

So proved

Tom.
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3. Let G be a group of order 56. 　　　　a.: 50

Prove that G is non-simple.

proof. Notice that $56 = 2^3 \cdot 7$, and denote n_p is number of sylow p-subgroups.
By the sylow theorem, $n_7 \in \{1, 8\}$ and $n_2 \in \{1, 7\}$.
Either $n_7 = 1$ or $n_2 = 1$, then the sylow subgroup is normal, so we are done. Suppose that $n_7 = 8$ and $n_2 = 7$.
by the Lagrange's theorem, all non-identity elements in sylow 7-subgroup has order of 7, Since $n_7 = 8$, there are $8 \cdot 6 = 48$ distinct elements of order 7 in G. Meanwhile, Since $n_2 = 7$, let $P, Q \in Syl_2(G)$, Then $|P \cap Q| \leq 4$, so there are $7 + 4 = 11$ distict elemones whose order is multiple of 2, but $48 + 11 = 59 > 56$, it contradicts. ∎

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4.

a)

b)

c)

Proof.

5.

a).

b).

c).

Proof. a).

6.

Proof.

7.

a)

b)

c)

9.

Proof.

11.

a)

b)

c).